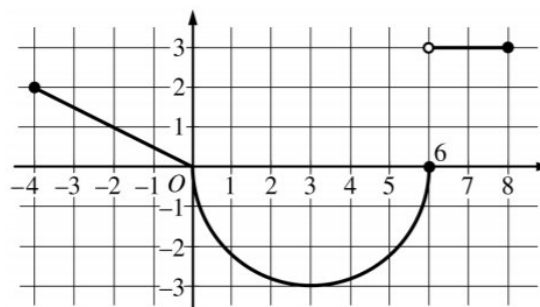


4-7-7

1. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of g

The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t) dt$.

- (a) Find $f(7)$ and $f'(7)$.
- (b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.
- (c) For each of $\lim_{x \rightarrow 0^-} g'(x)$ and $\lim_{x \rightarrow 0^+} g'(x)$, find the value or state that it does not exist.
- (d) Find $\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1}$.

Part A**1 point is earned for: $f(7)$** **1 point is earned for: $f'(7)$**

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $f(7)$ **1 point is earned for: $f'(7)$**

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$

4-7-7

Part B**2 points are earned for:** answer with justificationOn the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: answer with justificationOn the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.**Part C****1 point is earned for:** left-hand limit**1 point is earned for:** right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: left-hand limit**1 point is earned for:** right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

4-7-7

$\lim_{x \rightarrow 0^+} g'(x)$ does not exist.

Part D

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

4-7-7

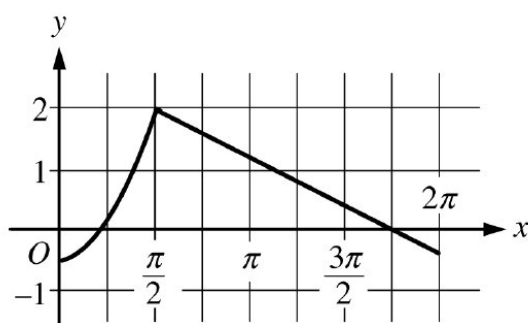
2. Let f be the function defined by $f(x) = e^x \cos(x)$.

(a) Find the average rate of change of f on the interval $0 \leq x \leq \pi$.

(b) What is the slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$?

(c) Find the absolute minimum value of f on the interval $0 \leq x \leq 2\pi$. Justify your answer.

(d) Let g be a differentiable function such that $g\left(\frac{\pi}{2}\right) = 0$. The graph of g' , the derivative of g , is shown below. Find the value of $\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)}$ or state that it does not exist. Justify your answer.



Graph of g'

Part A

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.



0

1

The student response earns one of the following points:

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.

Part B

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

4-7-7

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.

Part C

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ as candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

4-7-7

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies as $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

4-7-7

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.

Part D

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

4-7-7

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$

4-7-7

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Functions f , g , and h are twice-differentiable functions with $g(2) = h(2) = 4$. The line $y = 4 + \frac{2}{3}(x - 2)$ is tangent to both the graph of g at $x = 2$ and the graph of h at $x = 2$.

a) Find $h'(2)$.

b) Let a be the function given by $a(x) = 3x^3h(x)$. Write an expression for $a'(x)$. Find $a'(2)$.

c) The function h satisfies $h(x) = \frac{x^2 - 4}{1 - (f(x))^3}$ for $x \neq 2$. It is known that $\lim_{x \rightarrow 2} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x \rightarrow 2} h(x)$ to find $f(2)$ and $f'(2)$. Show the work that leads to your answers.

d) It is known that $g(x) \leq h(x)$ for $1 < x < 3$. Let k be a function satisfying $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$. Is k continuous at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes the criteria below.

☐ · answer

Solution:

$$h'(2) = \frac{2}{3}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-7-7

The first point is earned for a derivative in the form of the product rule, namely $a'(x) = p(x) \cdot h(x) + q(x) \cdot h'(x)$, where $p(x)$ and $q(x)$ are both polynomials.

Simplification is not required to earn the third point. Function values must be substituted into the expression.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ form of product rule
- ☐ $a'(x)$
- ☐ $a'(2)$

Solution:

$$a'(x) = 9x^2h(x) + 3x^3h'(x)$$

$$a'(2) = 9 \cdot 2^2h(2) + 3 \cdot 2^3h'(2) = 36 \cdot 4 + 24 \cdot \frac{2}{3} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for showing $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$. This work may appear anywhere in the response.

The second point is earned for calculating $f(2) = 0$ in the context of setting the denominator $1 - (f(x))^3$ equal to 0.

The third point is earned for evidence of applying L'Hospital's Rule, which includes a new limit of a quotient where differentiation was attempted in both the numerator and denominator. Note that incorrect notation such as $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = \frac{0}{0}$ makes a response not eligible to earn the third point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

☐ $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$

4-7-7

- ☐ $f(2)$
- ☐ L'Hospital's Rule
- ☐ $f'(2)$

Solution:

Because h is differentiable, h is continuous, so $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Also, $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3}$, so $\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = 4$.

Because $\lim_{x \rightarrow 2} (x^2 - 4) = 0$, we must also have $\lim_{x \rightarrow 2} (1 - (f(x))^3) = 0$.

Thus $\lim_{x \rightarrow 2} f(x) = 1$.

Because f is differentiable, f is continuous, so $f(2) = \lim_{x \rightarrow 2} f(x) = 1$.

Also, because f is twice differentiable, f' is continuous, so $\lim_{x \rightarrow 2} f'(x) = f'(2)$ exists.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = \lim_{x \rightarrow 2} \frac{2x}{-3(f(x))^2 f'(x)} = \frac{4}{-3(1)^2 \cdot f'(2)} = 4.$$

Thus $f'(2) = -\frac{1}{3}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point does not require naming the squeeze theorem. The justification requires (1) stating g and h are continuous, (2) stating $\lim_{x \rightarrow 2} g(x) = 4$ and $\lim_{x \rightarrow 2} h(x) = 4$ and concluding that $\lim_{x \rightarrow 2} k(x) = 4$, and (3) stating $g(2) = 4 = h(2)$ and concluding $k(2) = 4$.



0

1

The student response accurately includes the criteria below.

- ☐ continuous with justification

Solution:

4-7-7

Because g and h are differentiable, g and h are continuous, so $\lim_{x \rightarrow 2} g(x) = g(2) = 4$ and $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Because $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$, it follows from the squeeze theorem that $\lim_{x \rightarrow 2} k(x) = 4$.

Also, $4 = g(2) \leq k(2) \leq h(2) = 4$, so $k(2) = 4$.

Thus k is continuous at $x = 2$.

4-7-7

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x) dx$.

(d) Find $\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

4-7-7

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$

Part B**1 point is earned for:** integral bound**1 point is earned for:** answer with reasoning

Since $f'(x) \leq 7$, $\int_0^6 f'(x) dx \leq 6 \cdot 7 = 42$.

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral bound**1 point is earned for:** answer with reasoning

Since $f'(x) \leq 7$, $\int_0^6 f'(x) dx \leq 6 \cdot 7 = 42$.

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

Part C**1 point is earned for:** Fundamental Theorem of Calculus**1 point is earned for:** answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

4-7-7



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

4-7-7

5. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ is

- (A) ∞
(B) $e - 1$
(C) 1
(D) 0
(E) e^x



6. If a and b are positive constants, then $\lim_{x \rightarrow \infty} \frac{\ln(bx+1)}{\ln(ax^2+3)} =$

- (A) 0
(B) $\frac{1}{2}$
(C) $\frac{1}{2}ab$
(D) 2
(E) ∞



7. $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos x}$ is

- (A) -2
(B) 0
(C) 1
(D) 2
(E) nonexistent



8. $\lim_{x \rightarrow \infty} \frac{x^3}{e^{3x}}$ is

- (A) 0
(B) $\frac{2}{9}$
(C) $\frac{2}{3}$
(D) 1
(E) infinite



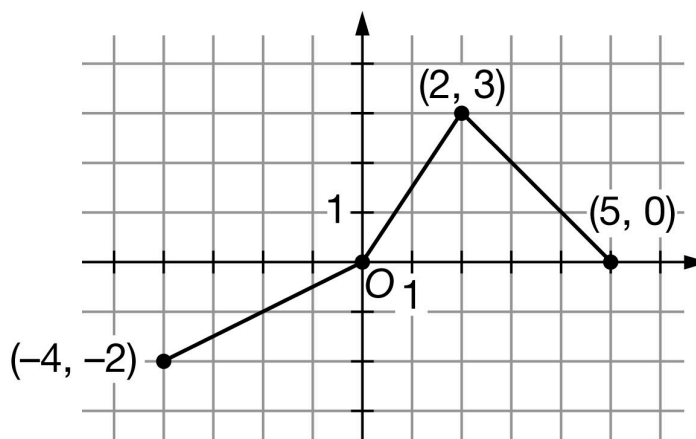
4-7-7

9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-4 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-4) = 7$.

- Find the x -coordinate of each critical point of g on the open interval $-4 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute maximum value of g on the interval $-4 \leq x \leq 5$. Justify your answer.
- Is $x = 2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = x \cdot g(x)$. It is known that $g(2) = 6$. Find the value of $h'(2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

4-7-7

- $x = 0$ and any additional x -coordinates not in $-4 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-4 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-4 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $g'(x)$ changes from negative to positive at $x = 0$
- the slope of g changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- g is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-4 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is NOT eligible to earn the point.

Note: If additional x -coordinates not in $-4 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for g because $g'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -4$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-4) = 7$ in a calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 10.5.

The ordered pair $(5, 10.5)$ is not a declaration of the maximum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-4	7
0	3
6	10.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

4-7-7

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-4 < x < 5$ is a relative minimum, so the absolute maximum value of g on $-4 \leq x \leq 5$ must occur at an endpoint.

$$g(-4) = 7$$

$$\begin{aligned} g(5) &= g(-4) + \int_{-4}^5 g'(t) \, dt \\ &= 7 + \int_{-4}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= 7 - \frac{1}{2} \cdot 4 \cdot 2 + \frac{1}{2} \cdot 3 \cdot 5 = 10.5 \end{aligned}$$

Therefore, the absolute maximum value of g on $-4 \leq x \leq 5$ is $g(5) = 10.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = 2$
- the slope of g' changes from positive to negative at $x = 2$
- g'' changes sign at $x = 2$
- a relative extrema exists on the graph of g' at $x = 2$
- a relative maximum exists on the graph of g' at $x = 2$

Does not earn the point:

- the concavity at $x = 2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ or $g''(x)$ does not exist (DNE)
- a relative minimum exists on the graph of g' at $x = 2$

4-7-7

- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = 2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is increasing for $x < 0$ and $x > 0$
- g' is increasing through $x = 0$
- g'' is always positive before and after $x = 0$
- the slope of g' is always positive before and after $x = 0$
- g'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always positive
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

Answer with reason

4-7-7

**Solution:**

$x = 0$ is not the location of a point of inflection for the graph of g because g' is increasing on both sides of $x = 0$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (g'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x)}{2x}$$

$$\cdot \frac{g'(-2)}{2(-2)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{8}$ or equivalent

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} g'(x) = g'(-2)$ and $\lim_{x \rightarrow -2} g''(x) = g''(-2)$.

$$\lim_{x \rightarrow -2} (g'(x) + 1) = g'(-2) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{g''(x)}{2x} \\ &= \frac{g''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{2}}{-4} = -\frac{1}{8} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product rule

Applies product rule: $h'(x) = x \cdot g'(x) + g(x)$

Product rule may appear as $h'(2) = 2 \cdot g'(2) + g(2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(2) = 12$ or equivalent

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = x \cdot g'(x) + g(x)$$

$$h'(2) = 2 \cdot g'(2) + g(2) = 2 \cdot 3 + 6 = 12$$

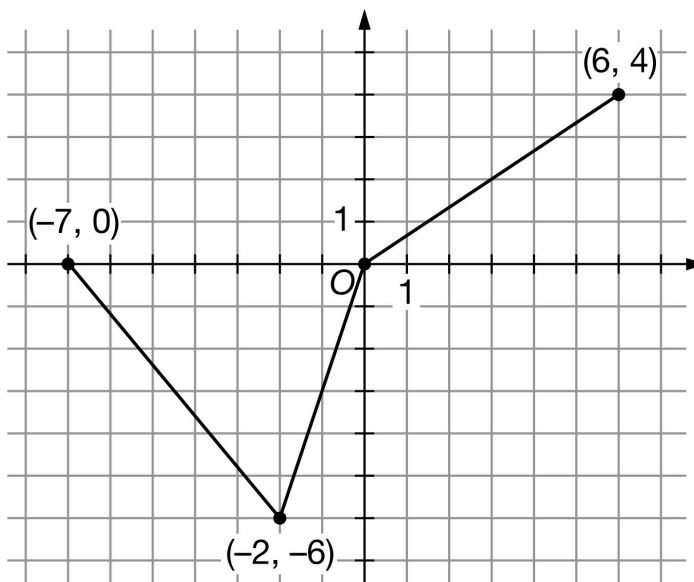
4-7-7

10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. It is known that $h(-7) = 2$.

- Find the x -coordinate of each critical point of h on the open interval $-7 < x < 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-7 \leq x \leq 6$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(-2) = -13$. Find the value of $g'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-7-7

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-7 < x < 6$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-7 < x < 6$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-7 < x < 6$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 0$
- the slope of h changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$
- Any incorrect argument

4-7-7

Note: If additional x -coordinates in $-7 < x < 6$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-7 < x < 6$ are identified in part (a), do not read reasons for these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -7$ and $x = 6$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-7) = 2$ in a calculation for $h(0)$ and/or $h(6)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 2.

The ordered pair $(-7, 2)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-7	2
0	-19
6	-7

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-7 < x < 6$ is a relative minimum, so the absolute maximum value of h on $-7 \leq x \leq 6$ must occur at an endpoint.

$$h(-7) = 2$$

$$\begin{aligned} h(6) &= h(-7) + \int_{-7}^6 h'(t) dt \\ &= 2 + \int_{-7}^0 h'(t) dt + \int_0^6 h'(t) dt \\ &= 2 - \frac{1}{2} \cdot 7 \cdot 6 + \frac{1}{2} \cdot 6 \cdot 4 = -7 \end{aligned}$$

Therefore, the absolute maximum value of h on $-7 \leq x \leq 6$ is $h(-7) = 2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' changes from negative to positive at $x = -2$
- the slope of h' changes from negative to positive at $x = -2$
- h'' changes sign at $x = -2$
- a relative extrema exists on the graph of h' at $x = -2$
- a relative minimum exists on the graph of h' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of h'']

4-7-7

- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = -2$
- The acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is increasing for $x < 0$ and $x > 0$
- h' is increasing through $x = 0$
- h'' is always positive before and after $x = 0$
- the slope of h' is always positive before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of h'']
- h' is always positive
- h' does not change sign

4-7-7



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 0$ is not the location a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} \rightarrow 0$

· $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow 3} \frac{2h'(x)}{2x}$

· $\frac{2h''(3)}{2(3)}$

· $\frac{2h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

4-7-7

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{2}{9}$ or equivalent

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:
 h' and h'' are continuous at $x = 3$. $\Rightarrow \lim_{x \rightarrow 3} h'(x) = h'(3)$ and $\lim_{x \rightarrow 3} h''(x) = h''(3)$.

$$\lim_{x \rightarrow 3} 2h'(x) - 4 = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \lim_{x \rightarrow 3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(3)}{3} = \frac{\frac{2}{3}}{3} = \frac{2}{9}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$ Product rule may appear as $g'(-2) = -2 \cdot h'(-2) + h(-2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

4-7-7

Eligibility: Earns the first point

Must be our answer, $g'(-2) = -1$ or equivalent

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(-2) = -2 \cdot h'(-2) + h(-2) = -2 \cdot (-6) + (-13) = -1$$

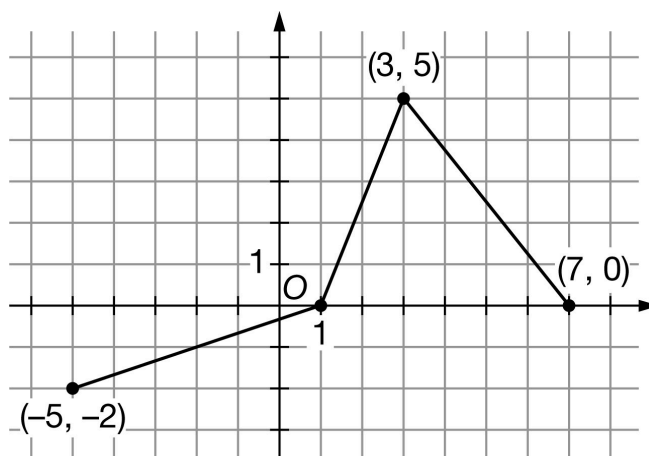
4-7-7

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-5 \leq x \leq 7$ consists of three line segments, as shown in the figure above. It is known that $h(-5) = 6$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 7$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 7$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 1$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(3) = 5$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 1$ only

4-7-7

- $x = 1$ and any additional x -coordinates not in $-5 < x < 7$

Does not earn the point:

- $x = 1$ and any additional x -coordinates in $-5 < x < 7$
- $x = 1$ with any incorrect supporting work

Do not read work related to additional values not in $-5 < x < 7$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$h'(x) = 0 \Rightarrow x = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 1$
- the slope of h changes from negative to positive at $x = 1$
- the graph changes from negative to positive at $x = 1$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 1$ and increasing to the right of $x = 1$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 1$

Note: If additional x -coordinates in $-5 < x < 7$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-5 < x < 7$ are identified in part (a), do not read these responses. Only read for values in the interval.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -5$ and $x = 7$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-5) = 6$ in calculation for $h(1)$ or $h(7)$

Third point: Answer with justification

The response must declare as an answer the absolute maximum value of 15.

The ordered pair $(7, 15)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-5	6
1	0
7	15



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

4-7-7

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-5 < x < 7$ is a relative minimum, so the absolute maximum value of h on $-5 \leq x \leq 7$ must occur at an endpoint.

$$h(-5) = 6$$

$$\begin{aligned} h(7) &= h(-5) + \int_{-5}^7 h'(t) \, dt \\ &= 6 + \int_{-5}^1 h'(t) \, dt + \int_1^7 h'(t) \, dt \\ &= 6 - \frac{1}{2} \cdot 6 \cdot 2 + \frac{1}{2} \cdot 6 \cdot 5 = 15 \end{aligned}$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 7$ is $h(7) = 15$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from increasing to decreasing
- h'' changes from positive to negative at $x = 3$
- the slope of h' changes from positive to negative at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative maximum exists on the graph of h' at $x = 2$

Does not earn the point:

- the concavity changes at $x = 3$ [Must appeal to the change in sign of h'']
- h' changes from positive to negative
- the graph changes from positive to negative
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of h' at $x = 2$

4-7-7

- the acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from increasing to decreasing there .

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 1$ is not the location of a point of inflection because

- h' is increasing for $x < 1$ and $x > 1$
- h' is increasing through $x = 1$
- h'' is always positive before and after $x = 1$
- the slope of h' is always positive before and after $x = 1$
- h'' does not change sign at $x = 1$
- a relative extrema does not exist on the graph of h' at $x = 1$

Does not earn the point:

- the concavity does not change [Must appeal to the change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

Answer with reason

4-7-7

**Solution:**

$x = 1$ is not the location of a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (h'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x)}{2x}$$

$$\cdot \frac{h'(-2)}{2(-2)}$$

$$\cdot \frac{h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{12}$ or equivalent

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$ and $\lim_{x \rightarrow -2} h''(x) = h''(-2)$.

$$\lim_{x \rightarrow -2} h'(x) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4} &= \lim_{x \rightarrow -2} \frac{h''(x)}{2x} \\ &= \frac{h''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{3}}{-4} = -\frac{1}{12} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

1st point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(3) = 3 \cdot h'(3) + h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the 1st point

Must be our answer, $g'(3) = 20$ or equivalent

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(3) = 3 \cdot h'(3) + h(3) = 3 \cdot 5 + 5 = 20$$

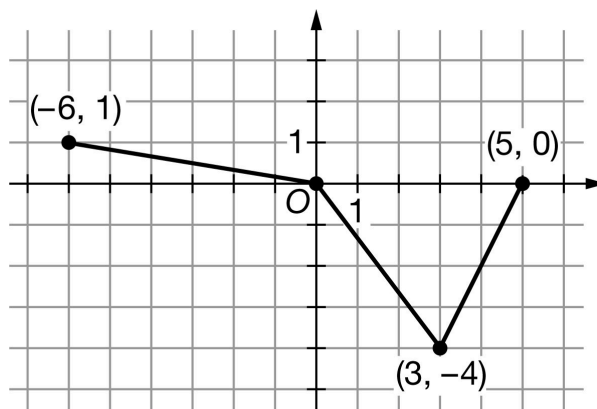
4-7-7

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $h(-6) = 5$.

- Find the x -coordinate of each critical point of h on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute minimum value of h on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -3} \frac{2h'(x)-1}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = 2x \cdot h(x)$. It is known that $h(3) = 2$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

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- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $h'(x)$ changes from positive to negative at $x = 0$
- the slope of h changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- h is increasing to the left of $x = 0$
- and decreasing to the right of $x = 0$
- (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to

4-7-7

earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for h because $h'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the 1st point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-6) = 5$ in a calculation for $h(0)$ or $h(5)$

Third point: Answer with justification

The response must declare as an answer the minimum value of -2 .

The ordered pair $(5, -2)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-6	5
0	8
5	-2



0	1	2	3
---	---	---	---

4-7-7

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of h on $-6 \leq x \leq 5$ must occur at an endpoint.

$$h(-6) = 5$$

$$\begin{aligned} h(5) &= h(-6) + \int_{-6}^5 h'(t) dt \\ &= 5 + \int_{-6}^0 h'(t) dt + \int_0^5 h'(t) dt \\ &= 5 + \frac{1}{2} \cdot 6 \cdot 1 - \frac{1}{2} \cdot 5 \cdot 4 = -2 \end{aligned}$$

Therefore, the absolute minimum value of h on $-6 \leq x \leq 5$ is $h(5) = -2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' shifts from negative to positive at $x = 3$
- the slope of h' shifts from negative to positive at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative minimum exists on the graph of h' at $x = 3$

Does not earn the point:

- the concavity at $x = 3$ changes. [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive

4-7-7

- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = 3$
- the acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is decreasing for $x < 0$ and $x > 0$
- h' is decreasing through $x = 0$
- h'' is always negative before and after $x = 0$
- the slope of h' is always negative before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change. [Must appeal to the lack of change in sign of h'']
- h' is always negative
- h' does not change sign



0	1
---	---

4-7-7

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of h because h' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -3$

Earns the point:

· $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} \rightarrow 0$

· $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -3} \frac{2h'(x)}{2x}$

· $\frac{2h'(-3)}{2(-3)}$

· $\frac{2h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

4-7-7

Must be our answer $\frac{1}{18}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -3$. $\Rightarrow \lim_{x \rightarrow -3} h'(x) = h'(-3)$ and $\lim_{x \rightarrow -3} h''(x) = h''(-3)$.

$$\lim_{x \rightarrow -3} 2h'(x) - 1 = 2 \cdot \frac{1}{2} - 1 = 0$$

$$\lim_{x \rightarrow -3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \lim_{x \rightarrow -3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(-3)}{-3} = \frac{-\frac{1}{6}}{-3} = \frac{1}{18}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = 2x \cdot h'(x) + 2 \cdot h(x)$

Product rule may appear as $g'(3) = 6 \cdot h'(3) + 2 \cdot h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

4-7-7

Must be our answer, $g'(3) = -20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$g'(x) = 2x \cdot h'(x) + 2h(x)$$

$$g'(3) = 2 \cdot 3 \cdot h'(3) + 2h(3) = 6 \cdot (-4) + 2 \cdot 2 = -20$$

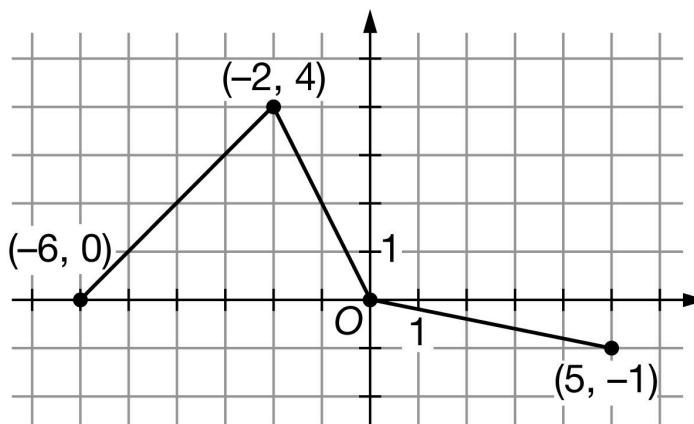
4-7-7

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-6) = -1$.

- Find the x -coordinate of each critical point of g on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute minimum value of g on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -1} \frac{g'(x)-2}{x^2-1}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = 3x \cdot g(x)$. It is known that $g(-2) = 7$. Find the value of $h'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

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- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $g'(x)$ changes from positive to negative at $x = 0$
- the slope of g changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- g is increasing to the left of $x = 0$ and decreasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for g because $g'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-6) = -1$ in calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the absolute minimum value of -1 .

The ordered pair $(-6, -1)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-6	-1
0	11
5	8.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

4-7-7

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of g on $-6 \leq x \leq 5$ must occur at an endpoint.

$$g(-6) = -1$$

$$\begin{aligned} g(5) &= g(-6) + \int_{-6}^5 g'(t) dt \\ &= -1 + \int_{-6}^0 g'(t) dt + \int_0^5 g'(t) dt \\ &= -1 + \frac{1}{2} \cdot 6 \cdot 4 - \frac{1}{2} \cdot 5 \cdot 1 = \frac{17}{2} \end{aligned}$$

Therefore, the absolute minimum value of g on $-6 \leq x \leq 5$ is $g(-6) = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = -2$
- the slope of g' changes from positive to negative at $x = -2$
- g'' changes sign at $x = -2$
- a relative extrema exists on the graph of g' at $x = -2$
- a relative maximum exists on the graph of g' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ OR $g''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of g' at $x = -2$

4-7-7

- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is decreasing for $x < 0$ and $x > 0$
- g' is decreasing through $x = 0$
- g'' is always negative before and after $x = 0$
- the slope of g' is always negative before and after $x = 0$
- g'' does not change sign at $x = 0$.
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always negative
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

4-7-7

☐ Answer with reason
Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -1$

Earns the point:

$$\cdot \lim_{x \rightarrow -1} (g'(x) - 2) = 0 \text{ and } \lim_{x \rightarrow -1} (x^2 - 1) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x)}{2x}$$

$$\cdot \frac{g''(-1)}{2(-1)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer 1 or equivalent

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -1$. $\Rightarrow \lim_{x \rightarrow -1} g'(x) = g'(-1)$ and $\lim_{x \rightarrow -1} g''(x) = g''(-1)$.

$$\lim_{x \rightarrow -1} g'(x) - 2 = 2 - 2 = 0$$

$$\lim_{x \rightarrow -1} (x^2 - 1) = 1^2 - 1 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{g''(x)}{2x} \\ &= \frac{g''(-1)}{2 \cdot (-1)} = \frac{-2}{-2} = 1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $h'(x) = 3x \cdot g'(x) + 3 \cdot g(x)$

Product rule may appear as $h'(-2) = -6 \cdot g'(-2) + 3 \cdot g(-2)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(-2) = -3$ or equivalent

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = 3x \cdot g'(x) + 3g(x)$$

$$h'(-2) = 3 \cdot (-2) \cdot g'(-2) + 3g(-2) = -6 \cdot 4 + 3 \cdot 7 = -3$$

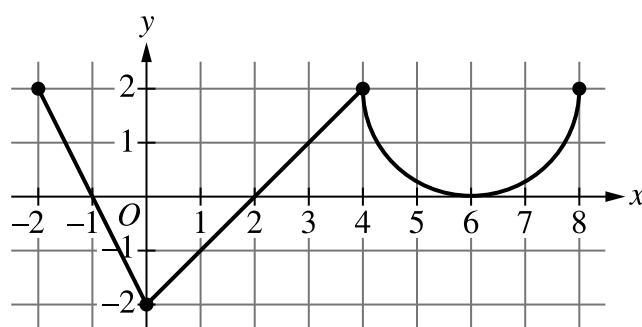
4-7-7

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The function f is defined on the closed interval $[-2, 8]$ and satisfies $f(2) = 1$. The graph of f' , the derivative of f , consists of two line segments and a semicircle, as shown in the figure.

- Does f have a relative minimum, a relative maximum, or neither at $x = 6$? Give a reason for your answer.
- On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.
- Find the value of $\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$, or show that it does not exist. Justify your answer.
- Find the absolute minimum value of f on the closed interval $[-2, 8]$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that declares $f'(x)$ does not change sign at $x = 6$, so neither, is sufficient to earn the point.

A response does not have to present intervals on which $f'(x)$ is positive or negative, but if any are given, they must be correct. Any presented intervals may include none, one, or both endpoints.

A response that declares $f'(x) > 0$ before and after $x = 6$ does not earn the point.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$f'(x) > 0$ on $(2, 6)$ and $f'(x) > 0$ on $(6, 8)$.

$f'(x)$ does not change sign at $x = 6$, so there is neither a relative maximum nor a relative minimum at this location.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only by an answer of $(-2, 0)$ and $(4, 6)$, or these intervals including one or both endpoints.

A response must earn the first point to be eligible for second point.

To earn the second point a response must correctly discuss the behavior of f' or the slopes of f' .

Special case: A response that presents exactly one of the two correct intervals with a correct reason earns 1 out of 2 points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Intervals

☐ Reason

Solution:

The graph of f is concave down on $(-2, 0)$ and $(4, 6)$ because f' is decreasing on these intervals.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-7-7

The first point is earned by the presentation of two separate limits for the numerator and denominator.

A response that presents a limit explicitly equal to $\frac{0}{0}$ does not earn the first point.

The second point is earned by applying L'Hospital's Rule, that is, by presenting at least one correct derivative in the limit of a ratio of derivatives.

The third point is earned for the correct answer with supporting work.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator
- ☐ Uses L'Hospital's Rule
- ☐ Answer

Solution:

Because f is differentiable at $x = 2$, f is continuous at $x = 2$, so $\lim_{x \rightarrow 2} f(x) = f(2) = 1$.

$$\lim_{x \rightarrow 2} (6f(x) - 3x) = 6 \cdot 1 - 3 \cdot 2 = 0$$

$$\lim_{x \rightarrow 2} (x^2 - 5x + 6) = 0$$

Because $\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$ is of indeterminate form $\frac{0}{0}$, L'Hospital's Rule can be applied.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6} = \lim_{x \rightarrow 2} \frac{6f'(x) - 3}{2x - 5} = \frac{6 \cdot 0 - 3}{2 \cdot 2 - 5} = 3.$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must state $f' = 0$ or equivalent. Listing the zeros of f' is not sufficient.

A response that presents any error in evaluating f at any critical point or endpoint will not earn the justification point.

A response need not present the value of $f(-1)$ provided $x = -1$ is eliminated because it is the location of a local maximum. A response need not present the value of $f(6)$ provided $x = 6$ is eliminated by reference to part (a) or eliminated through analysis.

4-7-7

A response need not present the value of $f(8)$ provided there is a presentation that argues $f'(x) \geq 0$ for $x > 2$ and, therefore, $f(8) > f(2)$.

A response that does not consider both endpoints does not earn the justification point.

The answer point is earned only for indicating that the minimum value is 1. It is not earned for noting that the minimum occurs at $x = 2$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
- ☐ Justification
- ☐ Answer

Solution:

$$f'(x) = 0 \Rightarrow x = -1, x = 2, x = 6$$

The function f is continuous on $[-2, 8]$, so the candidates for the location of an absolute minimum for f are $x = -2$, $x = -1$, $x = 2$, $x = 6$, and $x = 8$.

x	$f(x)$
-2	3
-1	4
2	1
6	$7 - \pi$
8	$11 - 2\pi$

The absolute minimum value of f is $f(2) = 1$.

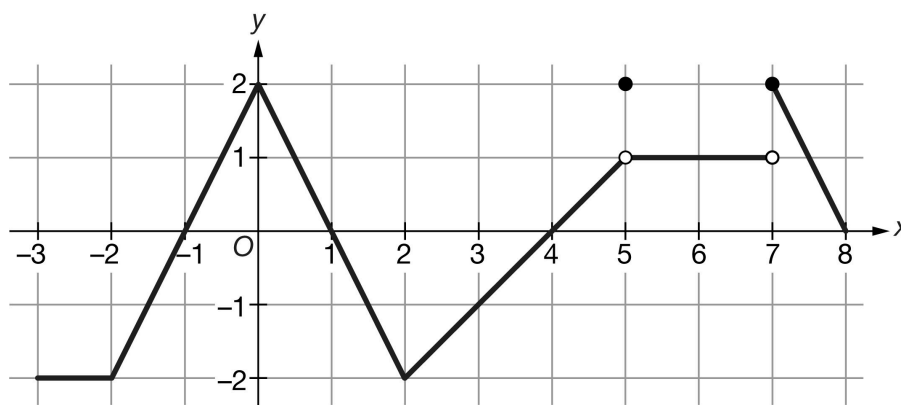
4-7-7

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the function f on the closed interval $-3 \leq x \leq 8$ consists of six line segments and the point $(5, 2)$, as shown in the figure above. The function g is given by

$$g(x) = \frac{1}{10}(4x^3 + 3x^2 - 10x - 17). \text{ It is known that } \int_{-3}^{-1} g(x) \, dx = -4.8 \text{ and } \int_{-3}^4 g(x) \, dx = 11.2.$$

(a) Find the value of $\int_4^8 f(x) \, dx$, or explain why the integral does not exist.

(b)

(i) Find the value of $\int_{-1}^4 g(x) \, dx$. Show the work that leads to your answer.

(ii) Find the value of $\int_{-1}^4 (2g(x) - 4f(x)) \, dx$. Show the work that leads to your answer.

(c) Let $h(x) = \begin{cases} g(x) & \text{for } x \leq -1 \\ f(x) + b & \text{for } x > -1 \end{cases}$. Find the value of b for which $\int_{-3}^4 h(x) \, dx = 14.2$.

(d) Find $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)+2}$. Show the work that leads to your answer.

Part A

The response should include supporting work for the numerical answer of 3.5.

4-7-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes the value of the integral.

Solution:

The integral exists since f is continuous on the interval $[4, 8]$, except for a removable discontinuity at $x = 5$ and a jump discontinuity at $x = 7$.

Using areas, $\int_4^8 f(x) dx = 0.5 + 2 + 1 = 3.5$.

Part B

Answers without supporting work do not earn any points.

A response that includes a sign error in an attempt to find $\int_{-1}^4 g(x) dx$ is eligible for the second and third points.

A response that includes a sign error in an attempt to find $\int_{-1}^4 f(x) dx$ is eligible for the second point and not eligible for the third point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\int_{-1}^4 g(x) dx$
- ☐ uses integral properties
- ☐ answer

Solution:

$$(i) \int_{-1}^4 g(x) dx = \int_{-3}^4 g(x) dx - \int_{-3}^{-1} g(x) dx = 11.2 - (-4.8) = 16$$

4-7-7

(ii) Using areas, $\int_{-1}^4 f(x) \, dx = 2 - 3 = -1$.

$$\begin{aligned} \int_{-1}^4 (2g(x) - 4f(x)) \, dx &= 2 \int_{-1}^4 g(x) \, dx - 4 \int_{-1}^4 f(x) \, dx \\ &= 2(16) - 4(-1) = 32 + 4 = 36 \end{aligned}$$

Part C

The evaluation of the definite integrals is assessed in the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses integral properties
- ☐ answer

Solution:

Using properties of integrals,

$$\begin{aligned} \int_{-3}^4 h(x) \, dx &= \int_{-3}^{-1} g(x) \, dx + \int_{-1}^4 (f(x) + b) \, dx \\ &= \int_{-3}^{-1} g(x) \, dx + \int_{-1}^4 f(x) \, dx + \int_{-1}^4 b \, dx \\ &= -4.8 + (-1) + (4 - (-1))b = -5.8 + 5b \\ -5.8 + 5b &= 14.2 \Rightarrow 5b = 20 \Rightarrow b = 4 \end{aligned}$$

Part D

For the first point, a response must show that $\lim_{x \rightarrow 1} f(x) = 0$ and $\lim_{x \rightarrow 1} (g(x) + 2) = 0$ and present an attempt at a limit of a ratio of derivatives. Use of $\frac{0}{0}$ in an equality statement with an expression or value is incorrect and results in the point not being earned.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-7-7



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ L'Hospital's Rule
- ☐ $g'(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 1} f(x) = 0 \text{ and } \lim_{x \rightarrow 1} (g(x) + 2) = -2 + 2 = 0.$$

Using L'Hospital's Rule,

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{f(x)}{g(x)+2} &= \lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 1} \frac{f'(x)}{\frac{1}{10}(12x^2+6x-10)} \\ &= \frac{-2}{\frac{1}{10}(12+6-10)} = \frac{-20}{8} = -\frac{5}{2}. \end{aligned}$$

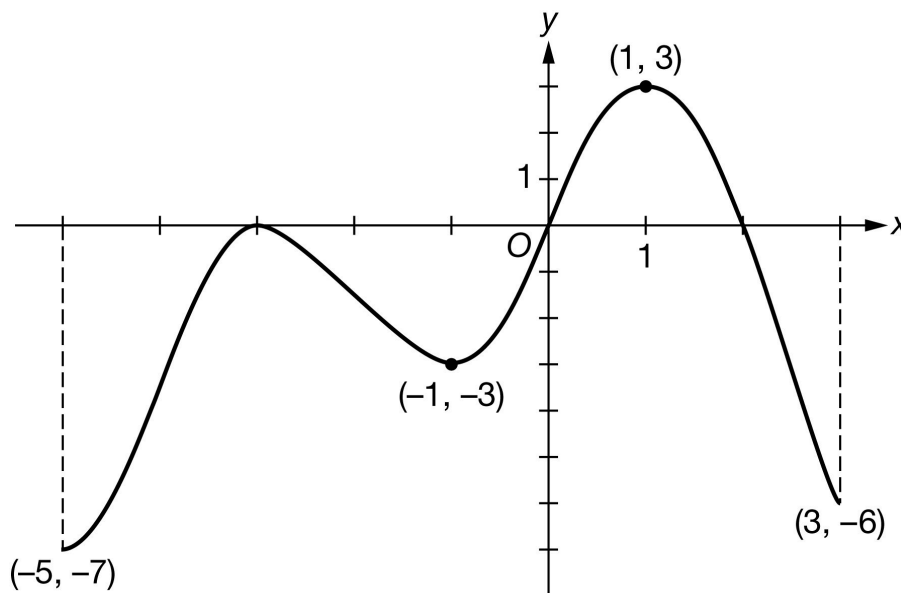
4-7-7

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-5 \leq x \leq 3$. The graph of h' has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 1$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-5, -3]$, $[-3, 0]$, $[0, 2]$, and $[2, 3]$ are 7, 5, 4, and 3, respectively. The function h is defined for all real numbers and satisfies $h(1) = 12$ and $h(3) = 11$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 3$.
- Identify the critical point from part (a) which corresponds to a relative maximum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of h on $-5 \leq x \leq 3$. Justify your answer.
- On what open intervals is h both increasing and concave down? Give a reason for your answer.
- Explain why there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.
- Find $\lim_{x \rightarrow 3^-} \frac{h(x) - 11}{x^2 - 9}$. Give a reason for your answer.

4-7-7

(h) The function p is defined by $p(x) = (h(x))^2$. Find $p'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 3)$. The point is still earned provided the three critical points, $x = -3$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(0, 0)$, $(2, 0)$ we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = -3$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as 2 and give the correct reason of “ h' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from increasing to decreasing at $x = 2$ ”
- Saying that “ $h'(2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

4-7-7

A response indicating that “ $h'(x)$ is positive for $x < 2$ and negative for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of h occurs where $h'(x)$ changes from positive to negative values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative maximum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -5$, $x = -3$, or $x = 0$ is selected as the location of the local maximum in part (b). Note that $h(-5) = 22$, $h(-3) = 15$, and $h(0) = 10$. No other x -coordinates will be read for consistency.

The response $h(2) = h(3) - (-3) = 14$ earns both points.

The response of both $11 - (-3) = 14$ and $h(2) = 14$ does not earn the first point, but earns the second point, provided no other work is presented.

4-7-7

Special Case: If a response uses $\int_1^2 h'(x) dx = 2$ then

$\cdot h(2) = h(1) + \int_1^2 h'(x) dx = 12 + 2 = 14$ earns both points, because it's possible to find

$\int_1^2 h'(x) dx = 2$ using the areas and initial conditions given.

$\cdot$ any response using perceived symmetry about $x = 1$ can earn at most the first point, for example

$h(2) = h(1) + \int_1^2 h'(x) dx = 12 + \frac{1}{2} \cdot 4 = 14$ earns just the first point.

Special Case: If in part (b), a response picks $x = 1$ or $x = 3$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(2) = h(3) + \int_3^2 h'(x) dx$

☐ Answer

Solution:

$$h(2) = h(3) + \int_3^2 h'(x) dx$$

$$= h(3) - \int_2^3 h'(x) dx = 11 - (-3) = 14$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x - value.

For $x = 2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

4-7-7

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $h(3) = 11$ by indicating that h' is negative on the interval $(2, 3)$, thus eliminating 11 as a possible absolute maximum.

A complete list of given or computable values for $h(x)$ is given in the table below

x	$h(x)$
-5	22
-3	15
0	10
1	12
2	14
3	11



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$\begin{aligned}
 h(-5) &= h(3) + \int_3^{-5} h'(x) \, dx \\
 &= h(3) - \int_{-5}^3 h'(x) \, dx = 11 - (-7 - 5 + 4 - 3) = 22
 \end{aligned}$$

$$h(2) = 14$$

$$h(3) = 11$$

So the absolute maximum value of h on $-5 \leq x \leq 3$ is $h(-5) = 22$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-7-7

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $[1, 2)$, $(1, 2]$, and $(1, 2)$ all earn the first point.

The reason “ h' is decreasing” can be expressed by either “ h'' is negative” or “the slope of h' is negative.”

We will overlook the false implication that $h' > 0$ on $[1, 2]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 2]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave down where h' is decreasing.

h is both increasing and concave down on the open interval $(1, 2)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 3]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 11 and 12, or mentioning $h(1)$ and $h(3)$

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 11.5 is between 11 and 12 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(3) = 11 < 11.5 < 12 = h(1)$$

By the Intermediate Value Theorem, there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(3) - 11 = 0$ and $3^2 - 9 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(3) = 11$ and $3^2 = 9$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h(3)-11}{3^2-9} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

4-7-7

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x} = \frac{-6}{6}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ and $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{6}{6}$ or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 3^-} h(x) = h(3)$ and $\lim_{x \rightarrow 3^-} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3^-} (h(x) - 11) = h(3) - 11 = 11 - 11 = 0$$

$$\lim_{x \rightarrow 3^-} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} &= \lim_{x \rightarrow 3^-} \frac{h'(x)}{2x} \\ &= \frac{h'(3)}{2 \cdot 3} = \frac{-6}{6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(1)$.

4-7-7

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(1)$.

All three points will be awarded for seeing only $2 \cdot 12 \cdot 3$, but $24 \cdot 3$, $2 \cdot 36$, or $6 \cdot 12$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$p'(x) = 2h(x) \cdot h'(x)$$

$$p'(1) = 2 \cdot h(1) \cdot h'(1) = 2 \cdot 12 \cdot 3 = 72$$

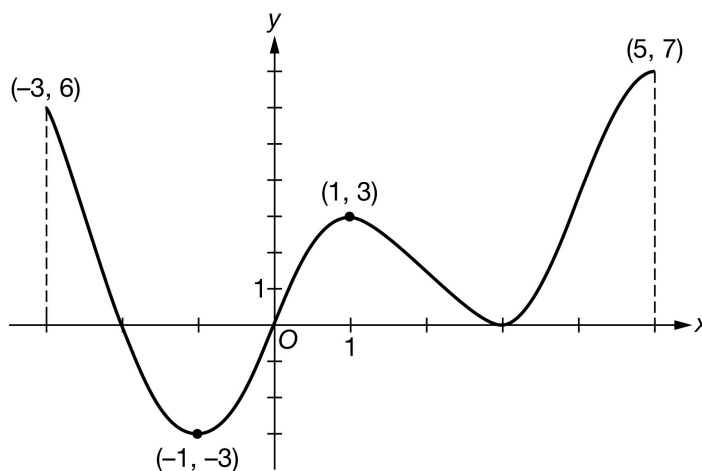
4-7-7

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -2]$, $[-2, 0]$, $[0, 3]$, and $[3, 5]$ are 3, 4, 5, and 7, respectively. The function g is defined for all real numbers and satisfies $g(1) = 3$ and $g(5) = 13$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative maximum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave up? Give a reason for your answer.
- Explain why there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15}$. Give a reason for your answer.
- The function r is defined by $r(x) = (g(x))^3$. Find $r'(1)$. Show the computations that lead to your answer.

Part A

4-7-7

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -2$, $x = 0$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(3, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -2$, $x = 0$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as -2 and give the correct reason of “ g' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from increasing to decreasing at $x = -2$ ”
- Saying that “ $g'(-2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is positive for $x < -2$ and negative for $x > -2$ ” is allowed to give this statement as a reason to earn the point.

4-7-7

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of g occurs where $g'(x)$ changes from positive to negative values.

$x = -2$ is the only location on $-3 < x < 5$ for which this is true, so $x = -2$ is the only location of a relative maximum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = 0$, or $x = 3$ is selected as the location of the local maximum in part (b). Note that $g(-3) = 2$, $g(0) = 1$, and $g(3) = 6$. No other x -coordinates will be read for consistency.

The response $g(-2) = g(5) - (-4 + 5 + 7) = 5$ earns both points.

The response of both $13 - 8 = 5$ and $g(-2) = 5$ does not earn the first point, but earns the second point, provided no other work is presented.

The string of equalities $g(-2) = g(5) - \int_{-2}^5 g'(x) dx = 13 - 8 = 5$ earns both points.

4-7-7

Special Case: If in part (b), a response picks $x = 1$ or $x = 5$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(-2) = g(5) + \int_5^{-2} g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(-2) &= g(5) + \int_5^{-2} g'(x) dx \\ &= g(5) - \int_{-2}^5 g'(x) dx = 13 - (7 + 5 - 4) = 5 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = -2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $g(-3) = 2$ by indicating that g' is positive on the interval $(-3, -2)$, thus eliminating 2 as a possible absolute maximum.

A complete list of given or computable values for $g(x)$ is given in the table below.

4-7-7

x	$g(x)$
-3	2
-2	5
0	1
1	3
3	6
5	13



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 13 - (7 + 5 - 4 + 3) = 2$$

$$g(-2) = 5$$

$$g(5) = 13$$

So the absolute maximum value of g on $-3 \leq x \leq 5$ is $g(5) = 13$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 0]$, $[-1, 0)$, $(-1, 0]$, $(-1, 0)$ all earn the first point.

The reason “ g' is increasing” can be expressed by either “ g'' is positive” or “the slope of g' is positive.”

We will overlook the false implication that $g' < 0$ on $[-1, 0]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 0]$.

4-7-7



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave up where g' is increasing.

g is both decreasing and concave up on the open interval $(-1, 0)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 3 and 13, or mentioning $g(1)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 10 is between 3 and 13 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

4-7-7

g is differentiable. $\Rightarrow g$ is continuous.

$$g(1) = 3 < 10 < 13 = g(5)$$

By the Intermediate Value Theorem, there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 13 = 0$ and $3 \cdot 5 - 15 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 13$ and $3 \cdot 5 = 15$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g(5)-13}{3 \cdot 5-15} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3} = \frac{7}{3}$, will earn the second point, but not the third point; note that even just one of two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{7}{3}$, or its numeric equivalent.

4-7-7



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 13) = g(5) - 13 = 13 - 13 = 0$$

$$\lim_{x \rightarrow 5^-} (3x - 15) = 3 \cdot 5 - 15 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{3} \\ &= \frac{g'(5)}{3} = \frac{7}{3} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(1))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(1)$.

All three points will be awarded for seeing only $3 \cdot 3^2 \cdot 3$, but $3 \cdot 9 \cdot 3$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

4-7-7

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$r'(x) = 3(g(x))^2 \cdot g'(x)$$

$$r'(1) = 3(g(1))^2 \cdot g'(1) = 3 \cdot 3^2 \cdot 3 = 81$$

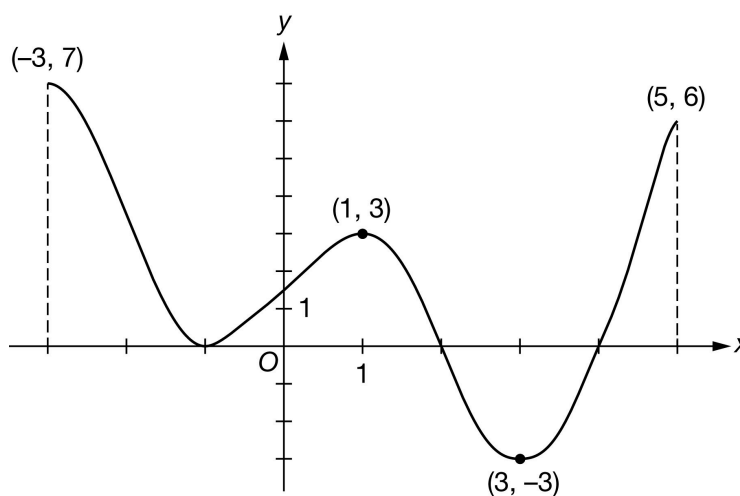
4-7-7

18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -3$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 7, 5, 4, and 3, respectively. The function g is defined for all real numbers and satisfies $g(3) = 8$ and $g(5) = 9$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25}$. Give a reason for your answer.
- The function f is defined by $f(x) = (g(x))^3$. Find $f'(3)$. Show the computations that lead to your answer.

4-7-7

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 4 and give the correct reason of “ g' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from decreasing to increasing at $x = 4$ ”
- Saying that “ $g'(4)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is negative for $x < 4$ and positive for $x > 4$ ” is allowed to give this statement as a reason to earn the point.

4-7-7

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the only location on $-3 < x < 5$ for which this is true, so $x = 4$ is the only location of a relative minimum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 2$ is selected as the location of the local minimum in part (b). Note that $g(-3) = -2$, $g(-1) = 5$, and $g(2) = 10$. No other x -coordinates will be read for consistency.

The response $g(4) = g(5) - 3 = 6$ earns both points.

The responses of both $9 - 3 = 6$ and $g(4) = 6$ do not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_4^3 g'(x) dx = 2$ then

$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - 2 = 6$ earns both points, because it's possible to find $\int_4^3 g'(x) dx = 2$ using the areas and initial conditions given.

4-7-7

any response using perceived symmetry about $x = 3$ can earn at most the first point, for example

$$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - \frac{1}{2} \cdot 4 = 6 \text{ earns just the first point.}$$

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(4) = g(5) + \int_5^4 g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(4) &= g(5) + \int_5^4 g'(x) dx \\ &= g(5) - \int_4^5 g'(x) dx = 9 - 3 = 6 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 4$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $g(5) = 9$ by indicating that g' is positive on the interval $(4, 5)$, thus eliminating 9 as a possible absolute minimum.

A complete list of given or computable values for $g(x)$ is given in the table below.

4-7-7

x	$g(x)$
-3	-2
1	5
2	10
3	8
4	6
5	9



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 9 - (7 + 5 - 4 + 3) = -2$$

$$g(4) = 6$$

$$g(5) = 9$$

So the absolute minimum value of g on $-3 \leq x \leq 5$ is $g(-3) = -2$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[2, 3]$, $[2, 3)$, $(2, 3]$, and $(2, 3)$ all earn the first point.

The reason “ g' is decreasing” can be expressed by either “ g'' is negative” or “the slope of g' is negative.”

We will overlook the false implication that $g' < 0$ on $[2, 3]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[2, 3]$.

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave down where g' is decreasing.

g is both decreasing and concave down on the open interval $(2, 3)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 8 and 9, or mentioning $g(3)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 8.5 is between 8 and 9 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

4-7-7

g is differentiable. $\Rightarrow g$ is continuous.

$$g(3) = 8 < 8.5 < 9 = g(5)$$

By the Intermediate Value Theorem, there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 9 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 9$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g(5)-9}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x} = \frac{3}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{5}$, or its numeric equivalent.

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ that both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 9) = g(5) - 9 = 9 - 9 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{2x} \\ &= \frac{g'(5)}{2 \cdot 5} = \frac{6}{10} = \frac{3}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(3))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(3)$.

All three points will be awarded for seeing only $3 \cdot 8^2 \cdot (-3)$, but $3 \cdot 64 \cdot (-3)$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

4-7-7

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$f'(x) = 3(g(x))^2 \cdot g'(x)$$

$$f'(3) = 3(g(3))^2 \cdot g'(3) = 3 \cdot 8^2 \cdot (-3) = -576$$

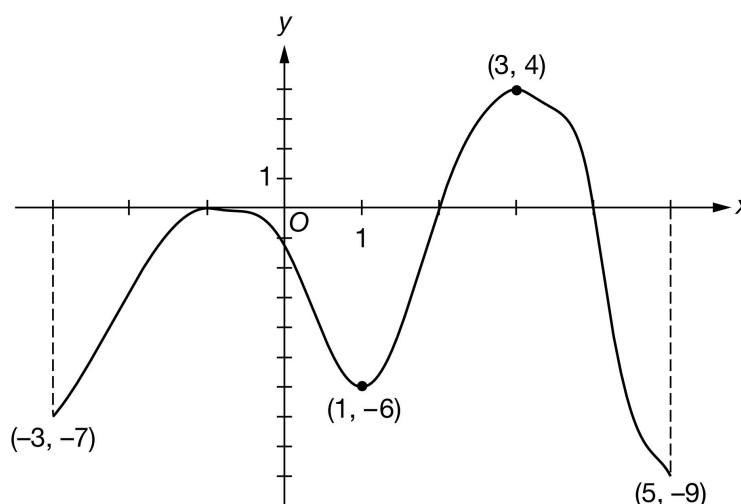
4-7-7

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $-3 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 6, 8, 5.6, and 6, respectively. The function f is defined for all real numbers and satisfies $f(3) = 10$ and $f(5) = 7$.

- Find the x -coordinate of each critical point of f on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of f . Give a reason for your answer.
- Find the value of f at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of f on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is f both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25}$. Give a reason for your answer.
- The function m is defined by $m(x) = (f(x))^2$. Find $m'(3)$. Show the computations that lead to your answer.

4-7-7

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$f'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 2 and give the correct reason of “ f' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ f switches from decreasing to increasing at $x = 2$ ”
- Saying that “ $f'(2)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of f' , the one given in the stem of the problem.

A response indicating that “ $f'(x)$ is negative for $x < 2$ and positive for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

4-7-7

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of f occurs where $f'(x)$ changes from negative to positive values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative minimum value of f on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $f(-3) = 21.4$, $f(-1) = 15.4$, and $f(4) = 13$. No other x -coordinates will be read for consistency.

The response $f(2) = f(5) - (5.6 - 6) = 7.4$ earns both points.

The responses of both $7 - (-0.4) = 7.4$ and $f(2) = 7.4$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $f(2) = f(5) - \int_2^5 f'(x) dx = 7 - (-0.4) = 7.4$ earns both points.

4-7-7

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f(2) = f(5) + \int_5^2 f'(x) dx$
- ☐ Answer

Solution:

$$\begin{aligned} f(2) &= f(5) + \int_5^2 f'(x) dx = f(5) - \int_2^5 f'(x) dx \\ &= f(5) - \int_2^4 f'(x) dx - \int_4^5 f'(x) dx = 7 - 5.6 - (-6) = 7.4 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $f(-3) = 21.4$ by indicating that f' is negative on the interval $(-3, -1)$, thus eliminating 21.4 as a possible absolute minimum.

A complete list of given or computable values for $f(x)$ is given in the table below.

4-7-7

x	$f(x)$
-3	21.4
-1	15.4
2	7.4
3	10
4	13
5	7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$f(-3) = f(5) + \int_5^{-3} f'(x) dx$$

$$= f(5) - \int_{-3}^5 f'(x) dx = 7 - (-6 + 5.6 - 8 - 6) = 21.4$$

$$f(2) = 7.4$$

$$f(5) = 7$$

So the absolute minimum value of f on $-3 \leq x \leq 5$ is

$$f(5) = 7.$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “ $(-1, 1]$ and $[4, 5)$ ” earns the first point.

The reason “ f' is decreasing” can be expressed by either “ f'' is negative” or “the slope of f' is negative.”

We will overlook the false implication that $f' < 0$ on $[-1, 1]$ and $[4, 5]$.

4-7-7

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 1] \cup [4, 5]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

f is decreasing where f' is negative.

f is concave down where f' is decreasing.

f is both decreasing and concave down on the open intervals $(-1, 1)$ and $(4, 5)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that f is continuous because f is differentiable or f is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of f need to be mentioned by either referring to the numerical values of 7 and 10, or mentioning $f(3)$ and $f(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since f is differentiable it’s continuous and 8.5 is between 7 and 10 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

4-7-7

Solution:

f is differentiable. $\Rightarrow f$ is continuous.

$$f(5) = 7 < 8.5 < 10 = f(3)$$

By the Intermediate Value Theorem, there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of f' or f .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $2f(5) - 14 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $2f(5) = 14$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f(5)-14}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x} = \frac{-9}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ and

$\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

4-7-7

The third point is earned only for the answer of $-\frac{9}{5}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

f is twice differentiable. $\Rightarrow$ both f' and f are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} f(x) = f(5)$ and $\lim_{x \rightarrow 5^-} f(x) = f(5)$.

$$\lim_{x \rightarrow 5^-} (2f(x) - 14) = 2 \cdot f(5) - 14 = 14 - 14 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 5 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{2f'(x)}{2x} \\ &= \frac{2f'(5)}{2 \cdot 5} = \frac{-9}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot f(x)$ or $2 \cdot f(3)$.

The chain rule point can be earned by seeing an expression involving f but not f' times $f'(3)$.

All three points will be awarded for seeing only $2 \cdot 10 \cdot 4$, but $20 \cdot 4$, $2 \cdot 40$, or $8 \cdot 10$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

4-7-7

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$m'(x) = 2f(x) \cdot f'(x)$$

$$m'(3) = 2 \cdot f(3) \cdot f'(3) = 2 \cdot 10 \cdot 4 = 80$$

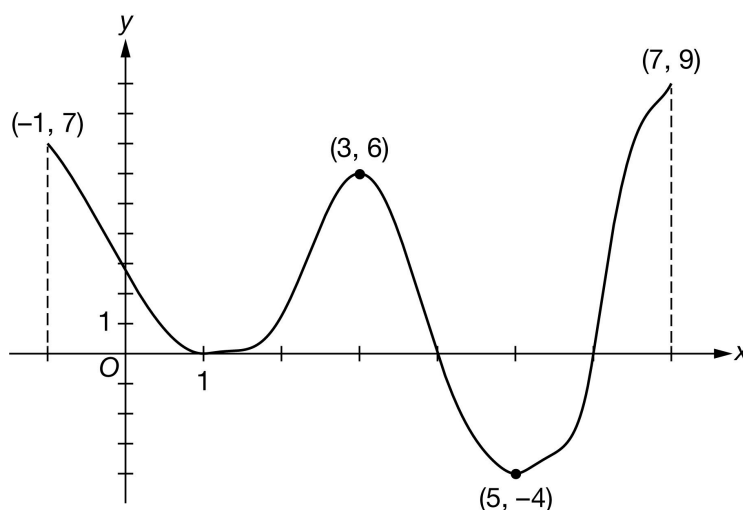
4-7-7

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-1 \leq x \leq 7$. The graph of h' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-1, 1]$, $[1, 4]$, $[4, 6]$, and $[6, 7]$ are 6, 8, 5.6, and 6, respectively. The function h is defined for all real numbers and satisfies $h(5) = 5$ and $h(7) = 8$.

- Find the x -coordinate of each critical point of h on the open interval $-1 < x < 7$.
- Identify the critical point from part (a) which corresponds to a relative minimum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of h on $-1 \leq x \leq 7$. Justify your answer.
- On what open intervals is h both increasing and concave up? Give a reason for your answer.
- Explain why there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.
- Find $\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21}$. Give a reason for your answer.
- The function a is defined by $a(x) = (h(x))^2$. Find $a'(5)$. Show the computations that lead to your answer.

4-7-7

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = 1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 7)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 7)$. The point is still earned provided the three critical points, $x = 1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(4, 0)$, $(6, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = 1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 6 and give the correct reason of “ h' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from decreasing to increasing at $x = 6$.”
- Saying that “ $h'(6)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

A response indicating that “ $h'(x)$ is negative for $x < 6$ and positive for $x > 6$ ” is allowed to give this statement as a reason to earn the point.

4-7-7

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of h occurs where $h'(x)$ changes from negative to positive values.

$x = 6$ is the only location on $-1 < x < 7$ for which this is true, so $x = 6$ is the only location of a relative minimum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -1$, $x = 1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $h(-1) = -6.4$, $h(1) = -0.4$, and $h(4) = 7.6$. No other x -coordinates will be read for consistency.

The response $h(6) = h(7) - 6 = 2$ earns both points.

The responses of both $8 - 6 = 2$ and $h(6) = 2$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $h(6) = h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2$ earns both points.

4-7-7

Special Case: If in part (b), a response picks $x = 5$ or $x = 7$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(6) = h(7) + \int_7^6 h'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} h(6) &= h(7) + \int_7^6 h'(x) dx \\ &= h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 6$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 5$), and thus the third point cannot be earned.

It is possible to not refer to $h(7) = 8$ by indicating that h' is positive on the interval $(6, 7)$, thus eliminating 8 as a possible absolute minimum.

A complete list of given or computable values for $h(x)$ is given in the table below

4-7-7

x	$h(x)$
-1	-6.4
1	-0.4
4	7.6
5	5
6	2
7	8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 6$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$h(-1) = h(7) + \int_7^{-1} h'(x) dx$$

$$= h(7) - \int_{-1}^7 h'(x) dx = 8 - (6 + 8 - 5.6 + 6) = -6.4$$

$$h(6) = 2$$

$$h(7) = 8$$

Therefore, the absolute minimum value of h on $-1 \leq x \leq 7$ is $h(-1) = -6.4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “(1, 3] and [6, 7)” earns the first point.

The reason “ h' is increasing” can be expressed by either “ h'' is positive” or “the slope of h' is positive.”

We will overlook the false implication that $h' > 0$ on $[1, 3]$ and $[6, 7]$.

4-7-7

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 3] \cup [6, 7]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave up where h' is increasing.

h is both increasing and concave up on the open intervals $(1, 3)$ and $(6, 7)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[5, 7]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 5 and 8, or mentioning $h(5)$ and $h(7)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 6.5 is between 5 and 8 such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

4-7-7

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(5) = 5 < 6.5 < 8 = h(7)$$

By the Intermediate Value Theorem, there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(7) - 8 = 0$ and $3 \cdot 7 - 21 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(7) = 8$ and $3 \cdot 7 = 21$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h(7)-8}{3 \cdot 7-21} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3} = \frac{9}{3}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ and

$\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

4-7-7

The third point is earned only for the answer of 3, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 7^-} h(x) = h(7)$ and $\lim_{x \rightarrow 7^-} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7^-} (h(x) - 8) = h(7) - 8 = 8 - 8 = 0$$

$$\lim_{x \rightarrow 7^-} (3x - 21) = 3 \cdot 7 - 21 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21} = \lim_{x \rightarrow 7^-} \frac{h'(x)}{3}$$

$$= \frac{h'(7)}{3} = \frac{9}{3} = 3$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The Power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(5)$.

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(5)$.

All three points will be awarded for seeing only $2 \cdot 5 \cdot (-4)$, but $10 \cdot (-4)$, $2 \cdot (-20)$, or $-8 \cdot 5$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

4-7-7

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$a'(x) = 2h(x) \cdot h'(x)$$

$$a'(5) = 2 \cdot h(5) \cdot h'(5) = 2 \cdot 5 \cdot (-4) = -40$$

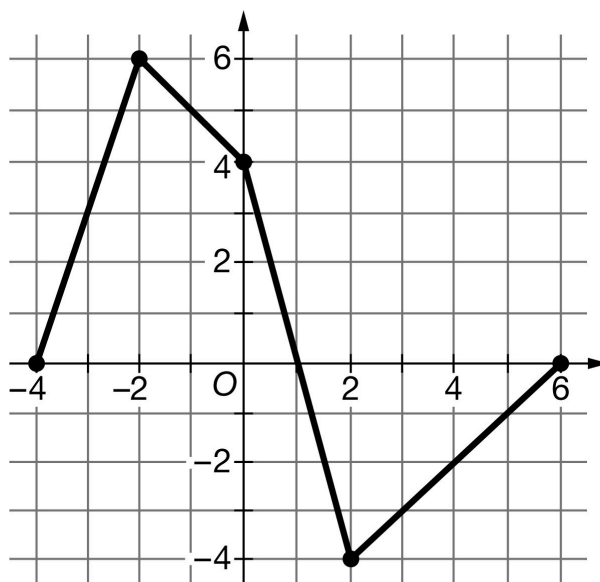
4-7-7

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be a continuous function defined on the closed interval $-4 \leq x \leq 6$. The graph of f , consisting of four line segments, is shown above. Let G be the function defined by $G(x) = \int_0^x f(t) \, dt$.

- On what open intervals is the graph of G concave up? Give a reason for your answer.
- Let P be the function defined by $P(x) = G(x) \cdot f(x)$. Find $P'(3)$.
- Find $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$.
- Find the average rate of change of G on the interval $[-4, 2]$. Does the Mean Value Theorem guarantee a value c , $-4 < c < 2$, for which $G'(c)$ is equal to this average rate of change? Justify your answer.

Global

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This “global point” can be earned in any one part. Expressions that show this connection and therefore earn this point include: $G' = f$, $G'(x) = f(x)$, $G''(x) = f'(x)$ in part (a), $G'(3) = f(3)$ in part (b), or $G'(2) = f(2)$

4-7-7

in part (c).



0	1
---	---

The student response accurately includes the criteria below.

☐ $G'(x) = f(x)$

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Intervals may also include one or both endpoints.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$$G'(x) = f(x)$$

The graph of G is concave up for $-4 < x < -2$ and $2 < x < 6$, because $G' = f$ is increasing on these intervals.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct application of the product rule in terms of x or in the evaluation of $P'(3)$. Once earned, this point cannot be lost.

The second point is earned by correctly evaluating $G(3) = -3.5$, $G'(3) = -3$, or $f(3) = -3$.

To be eligible to earn the third point a response must have earned the first two points.

Simplification of the numerical value is not required to earn the third point.

4-7-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ $G(3)$ or $G'(3)$
- ☐ Answer

Solution:

$$P'(x) = G(x) \cdot f'(x) + f(x) \cdot G'(x)$$

$$P'(3) = G(3) \cdot f'(3) + f(3) \cdot G'(3)$$

Substituting $G(3) = \int_0^3 f(t) dt = -3.5$ and $G'(3) = f(3) = -3$ into the above expression for $P'(3)$ gives the following:

$$P'(3) = -3.5 \cdot 1 + (-3) \cdot (-3) = 5.5$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must show $\lim_{x \rightarrow 2} (x^2 - 2x) = 0$ and $\lim_{x \rightarrow 2} G(x) = 0$ and must show a ratio of the two derivatives, $G'(x)$ and $2x - 2$. The ratio may be shown as evaluations of the derivatives at $x = 2$, such as $\frac{G'(2)}{2}$.

To earn the second point the response must evaluate correctly with appropriate limit notation. In particular the response must include $\lim_{x \rightarrow 2} \frac{G'(x)}{2x-2}$ or $\lim_{x \rightarrow 2} \frac{f(x)}{2x-2}$.

With any linkage errors (such as $\frac{G'(x)}{2x-2} = \frac{f(2)}{2}$), the response does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Uses L'Hospital's Rule

4-7-7

☐ Answer with justification
Solution:

$$\lim_{x \rightarrow 2} (x^2 - 2x) = 0$$

Because G is continuous for $-4 \leq x \leq 6$, $\lim_{x \rightarrow 2} G(x) = \int_0^2 f(t) dt = 0$.

Therefore, the limit $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$ is an indeterminate form of type $\frac{0}{0}$.

Using L'Hospital's Rule,

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{G'(x)}{2x - 2} \\ &= \lim_{x \rightarrow 2} \frac{f(x)}{2x - 2} = \frac{f(2)}{2} = \frac{-4}{2} = -2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present at least a difference and a quotient and a correct evaluation. For

example, $\frac{0+16}{6}$ or $\frac{G(2)-G(-4)}{6} = \frac{16}{6}$.

Simplification of the numerical value is not required to earn the first point, but any simplification must be correct.

The second point can be earned without the first point if the response has the correct setup but an incorrect or no evaluation of the average rate of change. The Mean Value Theorem need not be explicitly stated provided the conditions and conclusion are stated.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average rate of change
- ☐ Answer with justification

Solution:

$$G(2) = \int_0^2 f(t) dt = 0 \text{ and } G(-4) = \int_0^{-4} f(t) dt = -16$$

4-7-7

$$\text{Average rate of change} = \frac{G(2) - G(-4)}{2 - (-4)} = \frac{0 - (-16)}{6} = \frac{8}{3}$$

Yes, $G'(x) = f(x)$ so G is differentiable on $(-4, 2)$ and continuous on $[-4, 2]$. Therefore, the Mean Value Theorem applies and guarantees a value c , $-4 < c < 2$, such that $G'(c) = \frac{8}{3}$.

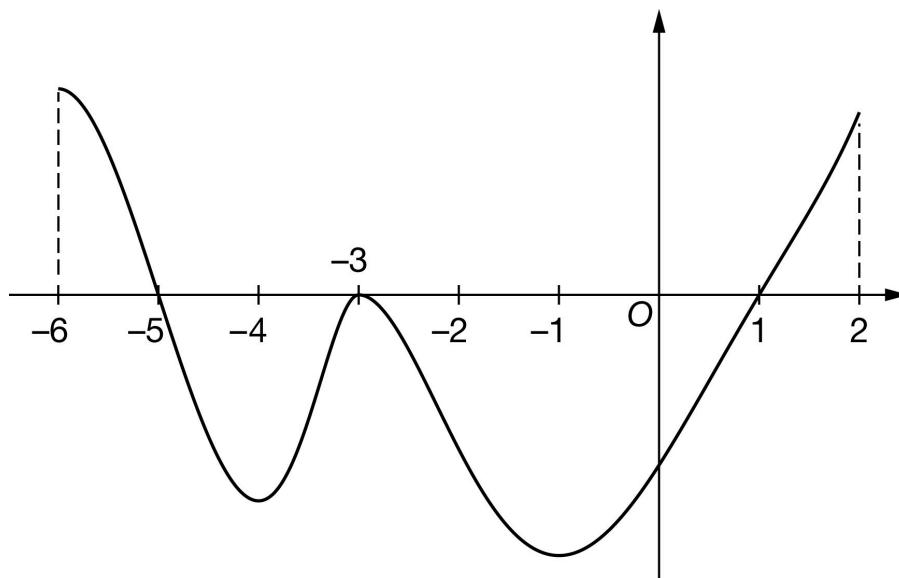
4-7-7

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-6 \leq x \leq 2$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-6, -5]$, $[-5, -3]$, $[-3, 1]$, and $[1, 2]$ are 9, 17, 42, and 6, respectively. The graph of g has horizontal tangents at $x = -4$, $x = -3$, and $x = -1$. Let h be the function defined by $h(x) = \int_{-3}^x g(t) dt$ for $-6 \leq x \leq 2$.

- Find the value of $h(1)$.
- Find the value of $h(-6)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-6 \leq x \leq 2$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-6 \leq x \leq 2$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.
- Find $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x}$. Give a reason for your answer.

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(h) The function p is defined by $p(x) = h(2x)$. It is known that $g(-2) = -10.8$. Find $p'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -42 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(1) = \int_{-3}^1 g(t) dt = -42$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 9 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 9 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-9 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-9 - 17$ earns the first point, but not the second point.
- The expression $-9 - (-17) + 42$ earns neither point.

Note: Just the list $-9, -17$ earns neither point. However, if the values -9 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals equal to 8 earns just the second point because not enough work has been shown.

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0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-6}^{-5} f(t) \, dt \right| = 9$ and $\left| \int_{-5}^{-3} f(t) \, dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} h(-6) &= \int_{-3}^{-6} g(t) \, dt = - \int_{-6}^{-3} g(t) \, dt \\ &= - \int_{-6}^{-5} g(t) \, dt - \int_{-5}^{-3} g(t) \, dt = -9 - (-17) = 8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -5$, $x = -3$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-6, 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-6, 2)$. The point is still earned provided the three critical points, $x = -5$, $x = -3$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5, 0)$, $(-3, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -5, x = -3, \text{ and } x = 1.$$

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -5$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-5)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -5$, $x = -3$, $x = 1$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-6) = 8$

- $h(-5) = 17$

- $h(-3) = 0$

- $h(1) = -42$

- $h(2) = -36$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -5$, h has a relative maximum because $h' = g$ is positive for $x < -5$ and $h' = g$ is negative for $x > -5$.
- At $x = -3$, h has neither because $h' = g$ is negative for $x < -3$ and $h' = g$ is negative for $x > -3$.
- At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -3$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list—4, —3, —1, and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

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· h' has a local extrema

Any incorrect statement (such as h' changes from increasing to decreasing at $x = -4$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -4$, $x = -3$, and $x = -1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow h$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-3 < x < 1$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $-42 < -33 < 0$, such a point exists” would earn the point.

The value of -33 must be bounded by -42 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-3)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $Ah(-3) = 0$.

A response may reference a value of $h(1)$ determined in part (a). If the value of $h(1)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -33 .

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0	1
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The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

$$h(1) = -42 < -33 < 0 = h(-3)$$

By the Intermediate Value Theorem, there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3 \cdot 1^2 - 3 = 0$ and $h(1) + 42 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3 \cdot 1^2 = 3$ and $h(1) = -42$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{0}{0}$ does not earn the first point. The statement

$\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} \rightarrow \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{6x}{h'(x) + 42}$, will earn the second point, but not the third point.

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Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{0}{0}$ and $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{6x}{h'(x)+42}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{1}{7}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 1} h(x) = h(1)$ and $\lim_{x \rightarrow 1} h'(x) = h'(1)$.

$$\lim_{x \rightarrow 1} (3x^2 - 3) = 3 \cdot 1^2 - 3 = 0$$

$$\lim_{x \rightarrow 1} (h(x) + 42x) = h(1) + 42 \cdot 1 = -42 + 42 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} &= \lim_{x \rightarrow 1} \frac{6x}{h'(x)+42} \\ &= \frac{6 \cdot 1}{h'(1)+42} = \frac{6}{g(1)+42} = \frac{6}{0+42} = \frac{1}{7} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(2x) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

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$$\cdot h'(2 \cdot (-1)) \cdot 2$$

$$\cdot g(2 \cdot (-1)) \cdot 2$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot 2$$

$$\cdot g(x) \cdot 2$$

$$\cdot h'(-2) \cdot 2$$

$$\cdot g(-2) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $p(x)$ or in $p'(-1)$. For example, the second point is demonstrated by $g(2x)$ or by $g(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $h'(2x) = g(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -10.8 . The function g does not have to be stated explicitly. For example, $h'(2 \cdot (-1)) \cdot 2 = -10.8 \cdot 2$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $g(2x) \cdot 2$ and $g(2 \cdot (-1)) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -21.6 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$p'(x) = h'(2x) \cdot 2$$

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$$= g(2x) \cdot 2$$

$$p'(-1) = g(2 \cdot (-1)) \cdot 2 = g(-2) \cdot 2 = (-10.8) \cdot 2 = -21.6$$

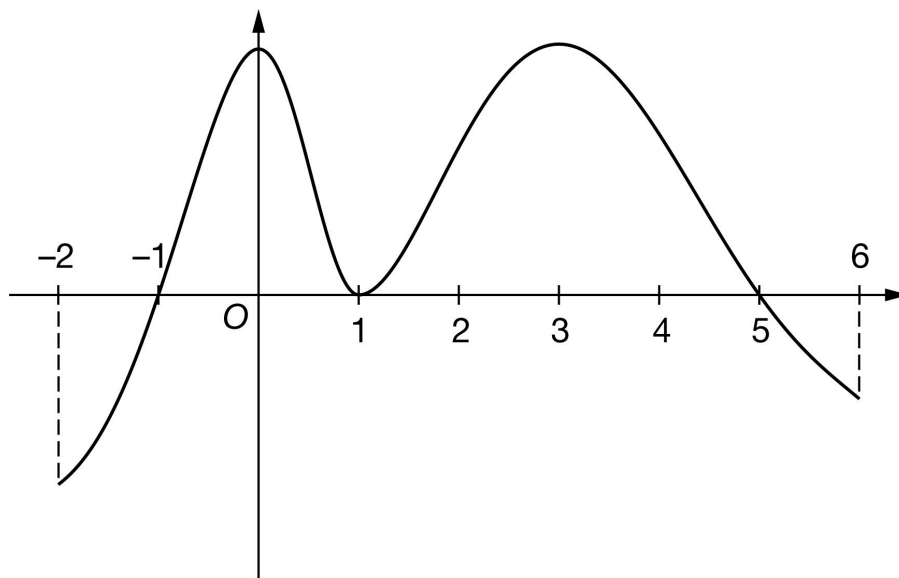
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23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the differentiable function h with domain $-2 \leq x \leq 6$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of h on the intervals $[-2, -1]$, $[-1, 1]$, $[1, 5]$, and $[5, 6]$ are 8, 19, 40, and 4, respectively. The graph of h has horizontal tangents at $x = 0$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_1^x h(t) \, dt$ for $-2 \leq x \leq 6$.

- (a) Find the value of $g(5)$.
- (b) Find the value of $g(-2)$. Show the computations that lead to your answer.
- (c) Find the x -coordinate of each critical point of g on the interval $-2 < x < 6$.
- (d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- (e) For $-2 < x < 6$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- (f) Explain why there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.
- (g) Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x}$. Give a reason for your answer.
- (h) The function z is defined by $z(x) = g(2x)$. It is known that $h(4) = 11.25$. Find $z'(2)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 40 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(5) = \int_1^5 h(t) dt = 40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 8 and ± 19 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 8 and ± 19 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-8) - 19$ earns both points because numerical expressions need not be simplified.
- The expression $(-8) - 19$ earns the first point, but not the second point.
- The expression $-(-8) - 19 - 40$ earns neither point.

Note: Just the list $-8, 19$ earns neither point. However, if the values -8 and 19 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -11 earns just the second point because not enough work has been shown.

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0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-2}^{-1} h(t) dt \right| = 8$ and $\left| \int_1^{-1} h(t) dt \right| = 19$

☐ Answer

Solution:

$$\begin{aligned} g(-2) &= \int_1^{-2} h(t) dt = - \int_{-2}^1 h(t) dt \\ &= - \int_{-2}^{-1} h(t) dt - \int_{-1}^1 h(t) dt = -(-8) - 19 = -11 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 1$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-2, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-2, 6)$. The point is still earned provided the three critical points, $x = -1$, $x = 1$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(5, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = h(x) = 0 \text{ at } x = -1, x = 1, \text{ and } x = 5.$$

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justification based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $h(x)$ and $g'(x)$ (that is, $h(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -1$, $x = 1$, $x = 5$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-2) = -11$

- $g(-1) = -19$

- $g(1) = 0$

- $g(5) = 40$

- $g(6) = 36$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -1$, g has a relative minimum because $g' = h$ is negative for $x < -1$ and $g' = h$ is positive for $x > -1$.
- At $x = 1$, g has neither because $g' = h$ is positive for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 1$, g has neither a relative minimum nor a relative maximum because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list 0, 1, 3, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $h(x)$ is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

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g' has a local extrema

Any incorrect statement (such as g' changes from decreasing to increasing at $x = 0$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = h$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = 0$, $x = 1$, and $x = 3$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that “ g is differentiable $\Rightarrow g$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 5$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $0 < 5 < 40$, such a point exists” would earn the point.

The value of 5 must be bounded by 0 and 40 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(1) = 0$ must be present.

A response may reference a value of $g(5)$ determined in part (a). If the value of $g(5)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 5.

4-7-7



0	1
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The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = h$ is differentiable on $1 \leq x \leq 5$. $\Rightarrow g$ is continuous on $1 \leq x \leq 5$.

$$g(1) = 0 < 5 < 40 = g(5)$$

By the Intermediate Value Theorem, there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $5^2 - 25 = 0$ and $g(5) - 40 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $5^2 = 25$ and $g(5) = 40$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$, will earn the second point, but not the third point.

4-7-7

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ and $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-5/4$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = h$ is differentiable. $\Rightarrow g' = h$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 5} g(x) = g(5)$ and $\lim_{x \rightarrow 5} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5} (x^2 - 25) = 5^2 - 25 = 0$$

$$\lim_{x \rightarrow 5} (g(x) - 8x) = g(5) - 8 \cdot 5 = 40 - 40 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} &= \lim_{x \rightarrow 5} \frac{2x}{g'(x) - 8} \\ &= \frac{2 \cdot 5}{g'(5) - 8} = \frac{10}{h(5) - 8} = \frac{10}{0 - 8} = -\frac{5}{4} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

4-7-7

$$\cdot g'(2 \cdot 2) \cdot 2$$

$$\cdot h(2 \cdot 2) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(2) \cdot 2$$

$$\cdot h(2) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking g' prime and h either directly or indirectly in the presented derivative of $z(x)$ or in $z'(2)$. For example, the second point is demonstrated by $h(2x)$ or by $h(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the Chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(4)$ to 11.25. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 2) \cdot 2 = 11.25 \cdot 2$ earns the first two points. Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 2) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of 22.5, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(2x) \cdot 2$$

4-7-7

$$= h(2x) \cdot 2$$

$$z'(2) = h(2 \cdot 2) \cdot 2 = h(4) \cdot 2 = 11.25 \cdot 2 = 22.5$$

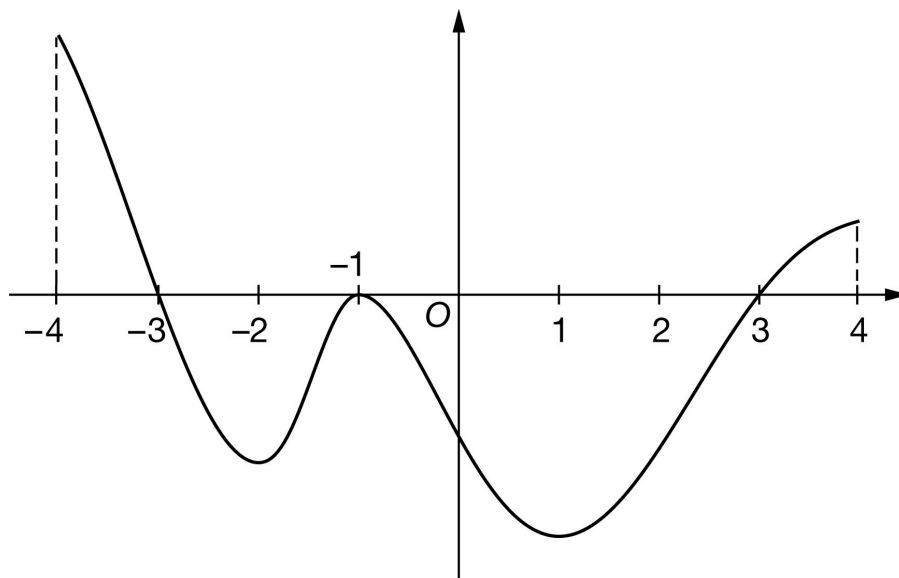
4-7-7

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 12, 17, 48, and 4, respectively. The graph of f has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_{-1}^x f(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $g(3)$.
- Find the value of $g(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of g on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- Explain why there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x}$. Give a reason for your answer.

4-7-7

(h) The function a is defined by $a(x) = g(x^3)$. It is known that $f(1) = -21.1$. Find $a'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -48 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3) = \int_{-1}^3 f(t) dt = -48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 12 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 12 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-12 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-12 - (-17)$ earns the first point, but not the second point.
- The expression $-12 - (-17) + 48$ earns neither point.

Note: Just the list $-12, -17$ earns neither point. However, if the values -12 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals 5 earns just the second point because not enough work has been shown.

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-3}^{-4} f(t) dt \right| = 12$ and $\left| \int_{-3}^{-1} f(t) dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} g(-4) &= \int_{-1}^{-4} f(t) dt = - \int_{-4}^{-1} f(t) dt \\ &= - \int_{-4}^{-3} f(t) dt - \int_{-3}^{-1} f(t) dt = -12 - (-17) = 5 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = f(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

4-7-7

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $g'(x)$ (that is, $f(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -3$, g has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-4) = 5$
- $g(-3) = 17$
- $g(-1) = 0$
- $g(3) = -48$
- $g(4) = -44$

4-7-7

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, g has a relative maximum because $g' = f$ is positive for $x < -3$ and $g' = f$ is negative for $x > -3$.
- At $x = -1$, g has neither because $g' = f$ is negative for $x < -1$ and $g' = f$ is negative for $x > -1$.
- At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = -1$, g has neither a relative minimum nor a relative maximum because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

4-7-7

g' has a local extrema

Any incorrect statement (such as g' changes from increasing to decreasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that g is differentiable $\Rightarrow g$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $-48 < -25 < 0$, such a point exists” would earn the point.

The value of -25 must be bounded by -48 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(-1) = 0$ must be present.

A response may reference a value of $g(3)$ determined in part (a). If the value of $g(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -25 .

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = f$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

$$g(3) = -48 < -25 < 0 = g(-1)$$

By the Intermediate Value Theorem, there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $g(3) + 48 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $g(3) = -48$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{2x}{g'(x) + 16}$, will earn the second point, but not the third point.

4-7-7

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} = \frac{2x}{g'(x)+16}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{8}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = f$ is differentiable. $\Rightarrow g' = f$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} g(x) = g(3)$ and $\lim_{x \rightarrow 3} g'(x) = g'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (g(x) + 16x) = g(3) + 16 \cdot 3 = -48 + 48 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{g(x)+16x} &= \lim_{x \rightarrow 3} \frac{2x}{g'(x)+16} \\ &= \frac{2 \cdot 3}{g'(3)+16} = \frac{6}{f(3)+16} = \frac{6}{0+16} = \frac{3}{8} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot g'(x^3) \cdot 3x^2$$

$$\cdot f(x^3) \cdot 3x^2$$

4-7-7

$$\cdot g'(1^3) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1^2$$

$$\cdot g'(1) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1$$

Responses that do not earn the point include:

$$\cdot g'(x^3) \cdot 3x$$

$$\cdot f(x^3) \cdot 3x$$

$$\cdot g'(x) \cdot 3x^2$$

$$\cdot f(x) \cdot 3x^2$$

$$\cdot g'(1) \cdot 3$$

$$\cdot f(1) \cdot 3$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $a(x)$ or in $a'(1)$. For example, the second point is earned by $f(x^3)$ or by $f(1^3)$.

The second point may be earned without earning the first point. For example, $g'(x^3) \cdot 3x = f(x^3) \cdot 3x$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(1)$ to -21.1 . The function f does not have to be stated explicitly. For example, $g'(1^3) \cdot 3 \cdot 1^2 = -21.1 \cdot 3 \cdot 1$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^3) \cdot 3x^2$ and $f(1^3) \cdot 3 \cdot 1^2$ earn the first two points.

The third point is earned for the correct answer of -63.3 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $g'(x)^3 \cdot 3x^2$ or $(g'(x))^3 \cdot 3x^2$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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4-7-7

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = g'(x^3) \cdot 3x^2$$

$$= f(x^3) \cdot 3x^2$$

$$a'(1) = g'(1^3) \cdot 3 \cdot 1^2 = g'(1) \cdot 3 = f(1) \cdot 3 = (-21.1) \cdot 3 = -63.3$$

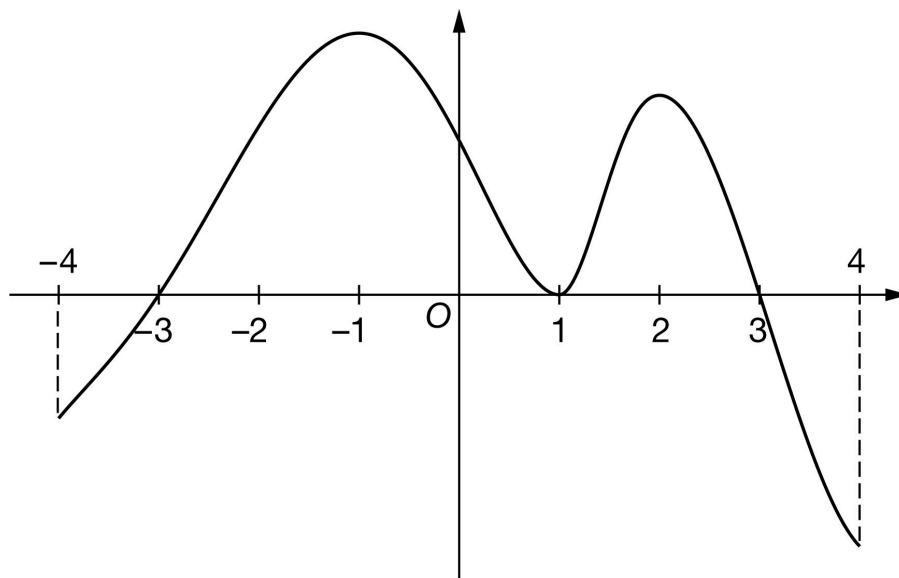
4-7-7

25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, 1]$, $[1, 3]$, and $[3, 4]$ are 5, 46, 18, and 11, respectively. The graph of f has horizontal tangents at $x = -1$, $x = 1$, and $x = 2$. Let h be the function defined by $h(x) = \int_1^x f(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $h(3)$.
- Find the value of $h(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x}$. Give a reason for your answer.

4-7-7

(h) The function m is defined by $m(x) = h(x^2)$. It is known that $f(4) = -19.3$. Find $m'(2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 18 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_1^3 f(t) dt = 18$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 5 and ± 46 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 5 and ± 46 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-5) - 46$ earns both points because numerical expressions need not be simplified.
- The expression $-5 - 46$ earns the first point, but not the second point.
- The expression $-(-5) - 46 + 8$ earns neither point.

Note: Just the list $-5, 46$ earns neither point. However, if the values -5 and 46 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -41 earns just the second point because not enough work has been shown.

4-7-7



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} f(t) \, dt \right| = 5$ and $\left| \int_{-3}^1 f(t) \, dt \right| = 46$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_1^{-4} f(t) \, dt = - \int_{-4}^1 f(t) \, dt \\ &= - \int_{-4}^{-3} f(t) \, dt - \int_{-3}^1 f(t) \, dt = -(-5) - 46 = -41 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = f(x) = 0 \text{ at } x = -3, x = 1, \text{ and } x = 3.$$

4-7-7

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $h'(x)$ (that is, $f(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = 1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-4) = -41$

- $h(-3) = -46$

- $h(1) = 0$

- $h(3) = 18$

- $h(4) = 7$

4-7-7

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.
- At $x = 1$, h has neither because $h' = f$ is positive for $x < 1$ and $h' = f$ is positive for $x > 1$.
- At $x = -3$, h has a relative maximum because $h' = f$ is positive for $x < 3$ and $h' = f$ is negative for $x > 3$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 1$, h has neither a relative minimum nor a relative maximum because $h'(x) = f(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-1, 1, 2$ and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

4-7-7

· h' has a local extrema

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -1$) fails to earn the second point.

If the response addresses at least one correct value of x it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -1$, $x = 1$, and $x = 2$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow$ that h is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 13 < 18$, such a point exists” would earn the point.

The value of 13 must be bounded by 0 and 18 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $h(1) = 0$ must be present.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 13.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = f$ is differentiable on $1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$$h(1) = 0 < 13 < 18 = h(3)$$

By the Intermediate Value Theorem, there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 18 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 18$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$, will earn the second point, but not the third point.

4-7-7

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{2x}{h'(x)-6}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of -1 , or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

$h' = f$ is differentiable. $\Rightarrow h' = f$ is continuous.

$\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 6x) = h(3) - 6 \cdot 3 = 18 - 18 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x)-6} \\ &= \frac{2 \cdot 3}{h'(3)-6} = \frac{6}{f(3)-6} = \frac{6}{0-6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

$$\cdot f(x^2) \cdot 2x$$

$$\cdot h'(2^2) \cdot 2 \cdot 2$$

4-7-7

$$\cdot f(2^2) \cdot 2 \cdot 2$$

$$\cdot h'(4) \cdot 2 \cdot 2$$

$$\cdot f(4) \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4$$

$$\cdot f(2^2) \cdot 4$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot f(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot f(x) \cdot 2x$$

$$\cdot h'(4) \cdot 4$$

$$\cdot f(4) \cdot 4$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $m(x)$ or in $m'(2)$. For example, the second point is earned by $f(x^2)$ or by $f(2^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(1)$ to -19.3 . The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'(2^2) \cdot 2 \cdot 2 = -19.3 \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4 = -19.3 \cdot 4$$

Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f(2^2) \cdot 2 \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -77.2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

4-7-7

Be careful of parenthesis errors in part (h). Responses of the form $h'(x^2) \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$m'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$m'(2) = f(2^2) \cdot 2 \cdot 2 = f(4) \cdot 4 = -19.3 \cdot 4 = -77.2$$

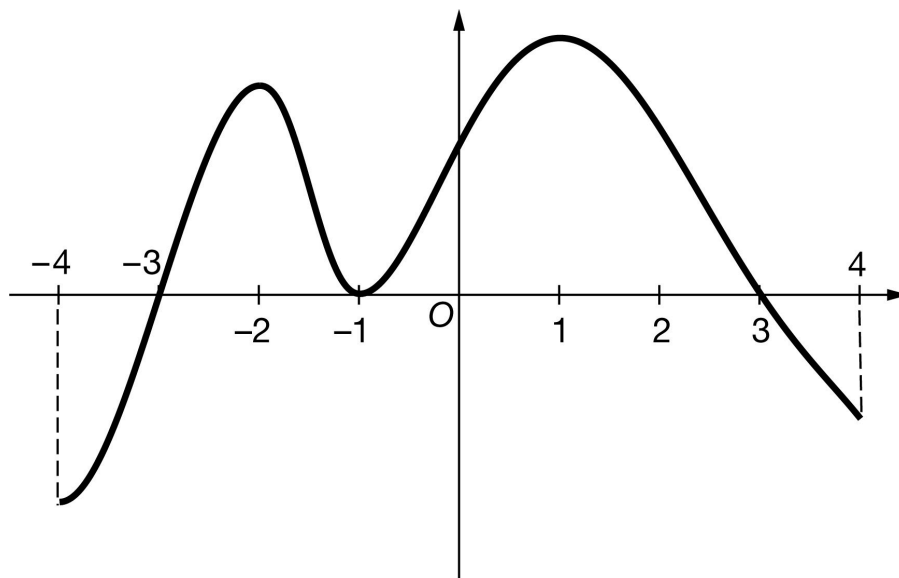
4-7-7

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 10, 18, 45, and 5, respectively. The graph of g has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let h be the function defined by $h(x) = \int_{-1}^x g(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $h(3)$.
- Find the value of $h(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x}$. Give a reason for your answer.

4-7-7

(h) The function k is defined by $k(x) = h(x^2)$. It is known that $g(1) = 19.7$. Find $k'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 45 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_{-1}^3 g(t) dt = 45$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 10 and ± 18 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 18 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-10) - 18$ earns both points because numerical expressions need not be simplified.
- The $(-10) - 18$ expression earns the first point, but not the second point.
- The expression $-(-10) - 18 - 45$ earns neither point.

Note: Just the list $-10, 18$ earns neither point. However, if the values -10 and 18 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals $= -8$ earns just the second point because not enough work has been shown.

4-7-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} g(t) \, dt \right| = 10$ and $\left| \int_{-3}^{-1} g(t) \, dt \right| = 18$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_{-1}^{-4} g(t) \, dt = - \int_{-4}^{-1} g(t) \, dt \\ &= - \int_{-4}^{-3} g(t) \, dt - \int_{-3}^{-1} g(t) \, dt = -(-10) - 18 = -8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$, read the x -coordinates and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

4-7-7

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $h(-4) = -8$

· $h(-3) = -18$

· $h(-1) = 0$

· $h(3) = 45$

· $h(4) = 40$

4-7-7

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.
- At $x = -1$, h has neither because $h' = g$ is positive for $x < -1$ and $h' = g$ is positive for $x > -1$.
- At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = -1$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- “ h' changes from increasing to decreasing or vice versa”
- “ h'' changes from positive to negative or vice versa”

4-7-7

· “ h' has a local extrema”

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that “ h is differentiable $\Rightarrow h$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 20 < 45$, such a point exists” would earn the point.

The value of 20 must be bounded by 0 and 45 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $h(-1) = 0$.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 20.

4-7-7



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $-1 \leq x \leq 3$.

$$h(-1) = 0 < 20 < 45 = h(3)$$

By the Intermediate Value Theorem, there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 45 = 0$.
- All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 45$ are not enough to earn the first point.

- The statement that the limit gives the indeterminate form $0/0$ earns the first point.
- Indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $0/0$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ does not earn the first point, but the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} \rightarrow \frac{0}{0}$ earns the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
- Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$, will earn the second point, but not the third point.

4-7-7

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-15x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-15x} = \frac{2x}{h'(x)-15}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{2}{5}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 15x) = h(3) - 15 \cdot 3 = 45 - 45 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-15x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x)-15} \\ &= \frac{2 \cdot 3}{h'(3)-15} = \frac{6}{g(3)-15} = \frac{6}{0-15} = -\frac{2}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

4-7-7

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot g((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot h'(1) \cdot 2 \cdot (-1)$$

$$\cdot g(1) \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2)$$

$$\cdot g((-1)^2) \cdot (-2)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(1) \cdot (-2)$$

$$\cdot g(1) \cdot (-2)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-1)$. For example, the second point is earned by $g(x^2)$ or by $g((-1)^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(1)$ to 19.7. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1) = 19.7 \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2) = 19.7 \cdot (-2)$$

4-7-7

Recall, if $x = -1$ is substituted for x , $x = -1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-1)^2) \cdot 2 \cdot (-1)$ earn the first two points.

The third point is earned for the correct answer of -39.4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-1) = g((-1)^2) \cdot 2 \cdot (-1) = g(1) \cdot (-2) = 19.7 \cdot (-2) = -39.4$$

27. $\lim_{x \rightarrow \infty} \frac{\ln(e^{3x} + x)}{x} =$

(A) 0

(B) 1

(C) 3

(D) ∞



4-7-7

28. $\lim_{t \rightarrow 0} \frac{\sin t}{\ln(2e^t - 1)} =$

(A) -1

(B) 0

(C) $\frac{1}{2}$

(D) 1

**Answer C**

Correct. Since $\lim_{t \rightarrow 0} \sin t = 0$ and $\lim_{t \rightarrow 0} \ln(2e^t - 1) = \ln(2 - 1) = 0$, the limit is an indeterminate form.

Therefore, L'Hospital's Rule can be used to find the limit.

$$\lim_{t \rightarrow 0} \frac{\sin t}{\ln(2e^t - 1)} = \lim_{t \rightarrow 0} \frac{\cos t}{\frac{1}{(2e^t - 1)} \cdot 2e^t} = \frac{1}{2}$$

29. Let f be the function defined by $f(x) = 3x + 2e^{-3x}$, and let g be a differentiable function with derivative given by $g'(x) = 4 + \frac{1}{x}$. It is known that $\lim_{x \rightarrow \infty} g(x) = \infty$. The value of $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ is

(A) 0

(B) $\frac{3}{4}$

(C) 1

(D) nonexistent

**Answer B**

Correct. Since $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow \infty} g(x) = \infty$, the limit is an indeterminate form. Therefore, L'Hospital's Rule can be used to find the value of the limit.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{3 - 6e^{-3x}}{4 + \frac{1}{x}} = \frac{3 - 0}{4} = \frac{3}{4}$$

4-7-7

30. Which of the following limits does not yield an indeterminate form?

(A) $\lim_{x \rightarrow 0} \frac{3x^2}{x - \sin x}$

(B) $\lim_{x \rightarrow 2} \frac{\ln(\frac{x}{2})}{x^2 - 5x + 6}$

(C) $\lim_{x \rightarrow \pi} \frac{x - \pi}{\cos x}$

(D) $\lim_{x \rightarrow \infty} \frac{e^{3x}}{x^{100}}$

**Answer C**

Correct. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is an indeterminate form. Here $\lim_{x \rightarrow \pi} (x - \pi) = 0$ but $\lim_{x \rightarrow \pi} \cos x = -1$. Therefore, $\lim_{x \rightarrow \pi} \frac{x - \pi}{\cos x}$ is not an indeterminate form and the value of the limit is 0.

31.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
3	2	-4	1	2

Selected values of the twice-differentiable functions f and g and their derivatives are given in the table above. The value of $\lim_{x \rightarrow 3} \frac{x^3 f(x) - 54}{g(x) - 1}$ is

(A) -108

(B) -54

(C) -27

(D) nonexistent

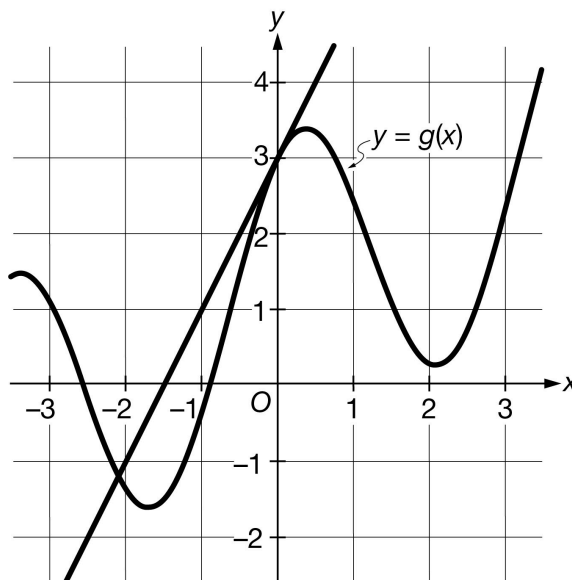
**Answer C**

Correct. Both f and g are continuous because they are differentiable. Since $\lim_{x \rightarrow 3} (x^3 f(x) - 54) = 27 \cdot f(3) - 54 = 27 \cdot 2 - 54 = 0$ and $\lim_{x \rightarrow 3} (g(x) - 1) = g(3) - 1 = 0$, the limit is of the indeterminate form $\frac{0}{0}$. Therefore, L'Hospital's Rule can be used to find the limit. Both f' and g' are continuous because they are differentiable.

$$\lim_{x \rightarrow 3} \frac{x^3 f(x) - 54}{g(x) - 1} = \lim_{x \rightarrow 3} \frac{3x^2 f(x) + f'(x)x^3}{g'(x)} = \frac{27 \cdot f(3) + f'(3) \cdot 27}{g'(3)} = \frac{27 \cdot 2 + (-4) \cdot 27}{2} = -\frac{54}{2} = -27$$

4-7-7

32.



The figure above shows the graph of the twice-differentiable function g and the line tangent to the graph of g at the point $(0, 3)$. The value of $\lim_{x \rightarrow 0} \frac{g(x)e^{-x}-3}{x^2-2x}$ is

(A) 1

(B) $\frac{1}{2}$ (C) $-\frac{5}{2}$

(D) nonexistent



Answer B

Correct. Substituting 0 into the fraction yields the indeterminate form $\frac{0}{0}$ because both the value of the numerator and the value of the denominator are equal to 0. From the graph, $g(0) = 3$. Therefore, L'Hospital's Rule can be used to find the limit.

$$\lim_{x \rightarrow 0} \frac{g(x)e^{-x}-3}{x^2-2x} = \lim_{x \rightarrow 0} \frac{g'(x)e^{-x}-g(x)e^{-x}}{2x-2}$$

Because $g(0) = 3$ and $g'(0) = 2$ from the graph, this limit evaluates as $\frac{2-3}{-2} = \frac{1}{2}$.

4-7-7

33. $\lim_{x \rightarrow \pi} \frac{x + \pi \sec x}{x^2 - \pi^2}$ is

(A) $-\frac{\pi}{2}$

(B) 0

(C) $\frac{1}{2\pi}$

(D) nonexistent

**Answer C**

Correct. Substituting π into the fraction yields the indeterminate form $\frac{0}{0}$ because both the value of the numerator and the value of the denominator are equal to 0. Therefore, L'Hospital's Rule can be used to find the limit.

$$\lim_{x \rightarrow \pi} \frac{x + \pi \sec x}{x^2 - \pi^2} = \lim_{x \rightarrow \pi} \frac{1 + \pi \sec x \tan x}{2x} = \frac{1}{2\pi}$$

34. $\lim_{x \rightarrow 0} \frac{6e^{4x} - 2e^{3x} - 4}{\sin(2x)} =$

(A) 2

(B) 4

(C) 9

(D) 18

**Answer C**

Correct. Since $\lim_{x \rightarrow 0} (6e^{4x} - 2e^{3x} - 4) = 6 - 2 - 4 = 0$ and $\lim_{x \rightarrow 0} \sin(2x) = \sin 0 = 0$, the limit is an indeterminate form. Therefore L'Hospital's Rule can be used to find the limit.

$$\lim_{x \rightarrow 0} \frac{6e^{4x} - 2e^{3x} - 4}{\sin(2x)} = \lim_{x \rightarrow 0} \frac{24e^{4x} - 6e^{3x}}{2 \cos(2x)} = \frac{24 - 6}{2} = 9$$

4-7-7

35. Let f be the function defined by $f(x) = 2x + 3e^{-5x}$, and let g be a differentiable function with derivative given by $g'(x) = \frac{1}{x} + 4\cos\left(\frac{5}{x}\right)$. It is known that $\lim_{x \rightarrow \infty} g(x) = \infty$. The value of $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ is
- (A) 0
- (B) $\frac{1}{2}$ ✓
- (C) 1
- (D) nonexistent

Answer B

Correct. Since $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow \infty} g(x) = \infty$, the limit is an indeterminate form. Therefore L'Hospital's Rule can be used to find the value of the limit.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{2 - 15e^{-5x}}{\frac{1}{x} + 4\cos\left(\frac{5}{x}\right)} = \frac{2 - 0}{0 + 4\cos 0} = \frac{2}{4} = \frac{1}{2}$$

36. Which of the following limits does not yield an indeterminate form?

- (A) $\lim_{x \rightarrow 0} \frac{4x^3}{\cos(x) - 1}$
- (B) $\lim_{x \rightarrow 3} \frac{\ln\left(\frac{x}{3}\right)}{x^2 - 7x + 12}$
- (C) $\lim_{x \rightarrow \pi} \frac{\pi - x}{\sin(2x) - 1}$ ✓
- (D) $\lim_{x \rightarrow \infty} \frac{x^{10}}{e^{2x} + x}$

Answer C

Correct. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is an indeterminate form.

Here $\lim_{x \rightarrow \pi} (\pi - x) = 0$ but $\lim_{x \rightarrow \pi} (\sin(2x) - 1) = -1$. Therefore, $\lim_{x \rightarrow \pi} \frac{\pi - x}{\sin(2x) - 1}$ is not an indeterminate form and the value of the limit is 0.

4-7-7

37.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	4	3	2	1

Selected values of the twice-differentiable functions f and g and their derivatives are given in the table above. The value of $\lim_{x \rightarrow 2} \frac{x^2 f(x) - 16}{g(x) - 2}$ is

(A) -28 (B) -12 (C) 28

(D) nonexistent

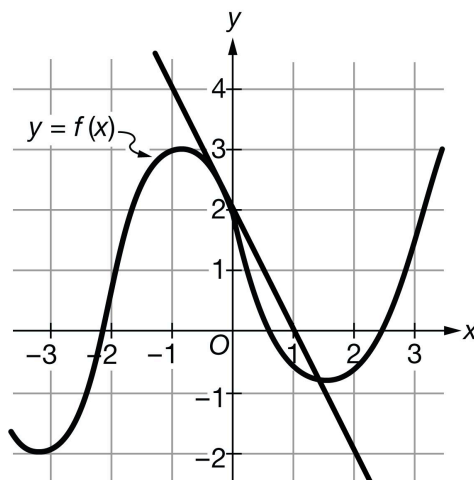
**Answer C**

Correct. Both f and g are continuous because they are differentiable. Since $\lim_{x \rightarrow 2} (x^2 f(x) - 16) = 4 \cdot f(2) - 16 = 4 \cdot 4 - 16 = 0$ and $\lim_{x \rightarrow 2} (g(x) - 2) = 2 - 2 = 0$, the limit is of the indeterminate form $\frac{0}{0}$. Therefore, L'Hospital's Rule can be used to find the limit. Both f' and g' are continuous because they are differentiable.

$$\lim_{x \rightarrow 2} \frac{x^2 f(x) - 16}{g(x) - 2} = \lim_{x \rightarrow 2} \frac{2x f(x) + f'(x) x^2}{g'(x)} = \frac{4 \cdot f(2) + f'(2) \cdot 4}{g'(2)} = \frac{4 \cdot 4 + 3 \cdot 4}{1} = 28$$

4-7-7

38.



The figure above shows the graph of the twice-differentiable function f and the line tangent to the graph of f at the point $(0, 2)$. The value of $\lim_{x \rightarrow 0} \frac{f(x)e^{-x} - 2}{x^2 - 2x}$ is

- (A) -1
- (B) 0
- (C) 2
- (D) nonexistent



Answer C

Correct. Substituting 0 into the fraction yields the indeterminate form $\frac{0}{0}$ because both the value of the numerator and the value of the denominator are equal to 0. From the graph, $f(0) = 2$. Therefore, L'Hospital's Rule can be used to find the limit.

$$\lim_{x \rightarrow 0} \frac{f(x)e^{-x} - 2}{x^2 - 2x} = \lim_{x \rightarrow 0} \frac{f'(x)e^{-x} - f(x)e^{-x}}{2x - 2}$$

Because $f(0) = 2$ and $f'(0) = -2$ from the graph, this limit evaluates as $\frac{-2 - 2}{-2} = 2$.

4-7-7

39. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{3 \cos x}{2x - \pi}$ is

(A) $-\frac{3}{2}$

(B) 0

(C) $\frac{3}{2}$

(D) nonexistent

**Answer A**

Correct. Substituting $\frac{\pi}{2}$ into the fraction yields the indeterminate form $\frac{0}{0}$ because both the value of the numerator and the value of the denominator are equal to 0. Therefore, L'Hospital's Rule can be used to find the limit.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{3 \cos x}{2x - \pi} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-3 \sin x}{2} = -\frac{3}{2}$$

40. $\lim_{x \rightarrow 0} \frac{4x^2}{e^{4x} - 4x - 1}$ is

(A) 0

(B) $1/2$

(C) 8

(D) non existent



4-7-7

41. $\lim_{x \rightarrow 3} \frac{\tan(x-3)}{3e^{x-3}-x}$ is

(A) 0

(B) $1/3$ (C) $1/2$

(D) nonexistent

**Answer C**

This option is correct. Substituting $x = 3$ into the fraction yields the indeterminate form $0/0$. Therefore L'Hospital's Rule can be used to find the limit:

$$\lim_{x \rightarrow 3} \frac{\tan(x-3)}{3e^{x-3}-x} = \lim_{x \rightarrow 3} \frac{\sec^2(x-3)}{3e^{x-3}-1} = \frac{1}{2}.$$

42. If $f(x) = \cos x$ and $g'(x) = 1$ for all x , and if $f(0) = g(0) = 0$, then $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ is

(A) $\frac{\pi}{2}$

(B) 1

(C) 0

(D) -1

(E) nonexistent



43. $\lim_{x \rightarrow 0} \frac{e^x - \cos x - 2x}{x^2 - 2x}$ is

(A) $-\frac{1}{2}$

(B) 0

(C) $\frac{1}{2}$

(D) 1

(E) nonexistent



4-7-7

44. $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{2 \sin^2 \theta}$

(A) 0

(B) 1/8

(C) 1/4



(D) 1

(E) nonexistent

45. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{\sin(\pi x)}$ is

(A) -2

(B) $-\frac{2}{\pi}$ 

(C) 0

(D) $\frac{2}{\pi}$

(E) nonexistent

46.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
2	0	0	5	7

The third derivative of the function f is continuous on the interval $(0, 4)$. Values for f and its first three derivatives at $x=2$ are given in the table above. What is $\lim_{x \rightarrow 2} \frac{f(x)}{(x-2)^2}$?

(A) 0

(B) $\frac{5}{2}$ 

(C) 5

(D) 7

(E) The limit does not exist.

47.

Let g be a continuously differentiable function with $g(1) = 6$ and $g'(1) = 3$. What is $\lim_{x \rightarrow 1} \frac{\int_1^x g(t) dt}{g(x) - 6}$?

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2



(E) The limit does not exist.

4-7-7

48. $\lim_{x \rightarrow 0} \frac{2x^6 + 6x^3}{4x^5 + 3x^3}$ is

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2



(E) nonexistent

49. $\lim_{x \rightarrow 0} \frac{5x^4 + 8x^2}{3x^4 - 16x^2}$ is

(A) $-\frac{1}{2}$ 

(B) 0

(C) 1

(D) $\frac{5}{3}$

(E) nonexistent

50. $\lim_{h \rightarrow 0} \frac{e^{(2+h)} - e^2}{h} =$

(A) 0

(B) 1

(C) $2e$ (D) e^2 (E) $2e^2$

51. $\lim_{h \rightarrow 0} \frac{e^h - 1}{2h}$ is

(A) 0

(B) $\frac{1}{2}$ 

(C) 1

(D) e

(E) nonexistent

4-7-7

52. $\lim_{x \rightarrow 0} \frac{\sin x \cos x}{x}$ is

(A) -1 (B) 0 (C) 1 (D) $\frac{\pi}{4}$

(E) nonexistent



53. The function g is continuous for all real numbers x and is defined by

$$g(x) = \frac{\cos(2x) - 1}{x^2} \text{ for } x \neq 0.$$

Use L'Hospital's Rule to find the value of $g(0)$. Show the work that leads to your answer.

Part A

One point is earned for uses L'Hospital's rule correctly at least once

One point is earned for the answer

$$\begin{aligned} g(0) &= \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{-2 \sin(2x)}{2x} \\ &= \lim_{x \rightarrow 0} \frac{-4 \cos(2x)}{2} = -2 \end{aligned}$$



0	1	2
---	---	---

The student earns all of the following points:

One point is earned for uses L'Hospital's rule correctly at least once

One point is earned for the answer

$$\begin{aligned} g(0) &= \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{-2 \sin(2x)}{2x} \\ &= \lim_{x \rightarrow 0} \frac{-4 \cos(2x)}{2} = -2 \end{aligned}$$

4-7-7

54. $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4}$ is

(A) $-\frac{1}{4}$

(B) 0

(C) 1

(D) $\frac{5}{4}$

(E) nonexistent



55.  Let f and g be functions that are differentiable for all real numbers, with $g(x) \neq 0$ for $x \neq 0$. If

$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} g(x) = 0$ and $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ exists, then $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ is

(A) 0

(B) $\frac{f'(x)}{g(x)}$

(C) $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$

(D) $\frac{f'(x)g(x) - f(x)g'(x)}{(f(x))^2}$

(E) nonexistent



4-7-7

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{17}{2} - \frac{3}{2} \cos\left(\frac{\pi t}{6}\right) & \text{for } 0 \leq t \leq 6 \\ 10 - \frac{1}{5}(t-6)^2 & \text{for } 6 < t \leq 10 \end{cases}$$

The depth of a river at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(8)$. Using correct units, explain the meaning of $W'(8)$ in the context of the problem.
- (b) The graph of W is concave down for $3 \leq t \leq 3.5$. Use the line tangent to the graph of W at $t = 3$ to show that $W(3.5) \leq 9$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(8)$

4-7-7

☐ interpretation with units
Solution:

For $6 < t < 10$, $W'(t) = -\frac{2}{5}(t - 6)$.

$$W'(8) = -\frac{2}{5}(8 - 6) = -\frac{4}{5}$$

At time $t = 8$ hours, the depth of the river is changing at a rate of $-\frac{4}{5}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(3)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(3) = \frac{17}{2} - \frac{3}{2}\cos\left(\frac{3\pi}{6}\right) = \frac{17}{2}$$

For $0 < t < 6$,

$$W'(t) = \frac{3}{2}\sin\left(\frac{\pi t}{6}\right) \cdot \frac{\pi}{6} = \frac{\pi}{4}\sin\left(\frac{\pi t}{6}\right).$$

$$W'(3) = \frac{\pi}{4}\sin\left(\frac{3\pi}{6}\right) = \frac{\pi}{4}$$

$$W(3.5) \approx \frac{17}{2} + \frac{\pi}{4}(3.5 - 3) = \frac{17}{2} + \frac{\pi}{8}$$

4-7-7

Because the graph of W is concave down for $3 \leq t \leq 3.5$, approximation of $W(3.5)$ is an overestimate of the actual value of $W(3.5)$.

$$\text{Therefore, } W(3.5) < \frac{17}{2} + \frac{\pi}{8} < \frac{17}{2} + \frac{1}{2} = 9.$$

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $= \frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\begin{aligned} \lim_{t \rightarrow 2} \left(W(t) - t^3 + \frac{1}{4} \right) &= W(2) - 2^3 + \frac{1}{4} \\ &= \frac{17}{2} - \frac{3}{2} \cos\left(\frac{2\pi}{6}\right) - 8 + \frac{1}{4} = 0 \end{aligned}$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\begin{aligned} \lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t - 2} &= \lim_{t \rightarrow 2} \frac{W'(t) - 3t^2}{1} = W'(2) - 3 \cdot 2^2 \\ &= \frac{\pi}{4} \sin\left(\frac{2\pi}{6}\right) - 12 = \frac{\pi\sqrt{3}}{8} - 12 \end{aligned}$$

4-7-7

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{32}{5} + \frac{2}{5}\cos\left(\frac{\pi t}{4}\right) & \text{for } 0 \leq t \leq 4 \\ 6 + \frac{1}{8}(t-4)^2 & \text{for } 4 < t \leq 9 \end{cases}$$

The depth of water in a zoo aquarium at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(7)$. Using correct units, explain the meaning of $W'(7)$ in the context of the problem.
- (b) The graph of W is concave up for $2 \leq t \leq 2.5$. Use the line tangent to the graph of W at $t = 2$ to show that $W(2.5) \geq 6$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(7)$

4-7-7

☐ interpretation with unit
Solution:

For $4 < t < 9$, $W'(t) = \frac{1}{4}(t - 4)$.

$$W'(7) = \frac{1}{4}(7 - 4) = \frac{3}{4}$$

At time $t = 7$ hours, the depth of the water in the zoo aquarium is changing at a rate of $\frac{3}{4}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(2)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(2) = \frac{32}{5} + \frac{2}{5}\cos\left(\frac{2\pi}{4}\right) = \frac{32}{5}$$

For $0 < t < 4$,

$$W'(t) = -\frac{2}{5}\sin\left(\frac{\pi t}{4}\right) \cdot \frac{\pi}{4} = -\frac{\pi}{10}\sin\left(\frac{\pi t}{4}\right).$$

$$W'(2) = -\frac{\pi}{10}\sin\left(\frac{2\pi}{4}\right) = -\frac{\pi}{10}$$

$$W(2.5) \approx \frac{32}{5} - \frac{\pi}{10}(2.5 - 2) = \frac{32}{5} - \frac{\pi}{20}$$

4-7-7

Because the graph of W is concave up for $2 \leq t \leq 2.5$, the tangent line approximation of $W(2.5)$ is an underestimate of the actual value of $W(2.5)$. Therefore, $W(2.5) > \frac{32}{5} - \frac{\pi}{20} > 6$.

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\begin{aligned}\lim_{t \rightarrow 2} \left(W(t) - t^2 - \frac{12}{5} \right) &= W(2) - 2^2 - \frac{12}{5} \\ &= \frac{32}{5} + \frac{2}{5} \cos\left(\frac{2\pi}{4}\right) - 4 - \frac{12}{5} = 0\end{aligned}$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\begin{aligned}\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t - 2} &= \lim_{t \rightarrow 2} \frac{W'(t) - 2t}{1} = W'(2) - 2 \cdot 2 \\ &= -\frac{\pi}{10} \sin\left(\frac{2\pi}{4}\right) - 4 = -\frac{\pi}{10} - 4\end{aligned}$$

4-7-7

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the y -axis so that its position at time t is given by $y(t) = 4t - \frac{2}{3}$ for all times t . A second particle, particle Q , moves along the x -axis so that its position at time t is given by $x(t) = \frac{\sin(\pi t)}{2-t}$ for all times $t \neq 2$.

- As time t approaches 2, what is the limit of the position of particle Q ? Show the work that leads to your answer.
- Show that the velocity of particle Q is given by $v_Q(t) = \frac{2\pi \cos(\pi t) - \pi t \cos(\pi t) + \sin(\pi t)}{(2-t)^2}$ for all times $t \neq 2$.
- Find the rate of change of the distance between particle P and particle Q at time $t = \frac{1}{2}$. Show the work that leads to your answer.

Part A

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $= \frac{0}{0}$ " or not using limit notation) impacts the second point.

The second point does not require a simplified answer. Trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts this point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ application of L'Hospital's Rule
- ☐ answer

Solution:

4-7-7

$$\lim_{t \rightarrow 2} \sin(\pi t) = 0$$

$$\lim_{t \rightarrow 2} (2 - t) = 0$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow 2} \frac{\sin(\pi t)}{2 - t} = \lim_{t \rightarrow 2} \frac{\pi \cos(\pi t)}{-1} = -\pi$$

Part B

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 2 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error.

The third point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ verification

Solution:

$$v_Q(t) = x'(t) = \frac{(2-t)\pi \cos(\pi t) - (-1)\sin(\pi t)}{(2-t)^2} = \frac{2\pi \cos(\pi t) - \pi t \cos(\pi t) + \sin(\pi t)}{(2-t)^2}$$

Part C

For the second point, trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

4-7-7

The student response accurately includes all four of the criteria below.

- ☐ $y' \left(\frac{1}{2} \right)$
- ☐ $x' \left(\frac{1}{2} \right)$
- ☐ related rates equation $2D \frac{dD}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$
- ☐ answer

At time $t = \frac{1}{2}$, particle P has position $y \left(\frac{1}{2} \right) = 4 \left(\frac{1}{2} \right) - \frac{2}{3} = 2 - \frac{2}{3} = \frac{4}{3}$ and velocity $y' \left(\frac{1}{2} \right) = v_P \left(\frac{1}{2} \right) = 4$.

At time $t = \frac{1}{2}$, particle Q has position $x \left(\frac{1}{2} \right) = \frac{\sin \left(\frac{\pi}{2} \right)}{2 - \frac{1}{2}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$ and velocity

$$x' \left(\frac{1}{2} \right) = v_Q \left(\frac{1}{2} \right) = \frac{2\pi \cos \left(\frac{\pi}{2} \right) - \pi \left(\frac{1}{2} \right) \cos \left(\frac{\pi}{2} \right) + \sin \left(\frac{\pi}{2} \right)}{\left(2 - \frac{1}{2} \right)^2} = \frac{1}{\left(\frac{3}{2} \right)^2} = \frac{4}{9}.$$

Let D represent the distance between particle P and particle Q . Using the Pythagorean Theorem, $D^2 = x^2 + y^2$.

$$\text{Therefore, } 2D \frac{dD}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}.$$

$$\frac{dD}{dt} = \frac{2x \frac{dx}{dt} + 2y \frac{dy}{dt}}{2D} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{D}$$

At time $t = \frac{1}{2}$,

$$D^2 = \left(\frac{2}{3} \right)^2 + \left(\frac{4}{3} \right)^2 \Rightarrow D^2 = \frac{20}{9} \Rightarrow D = \frac{\sqrt{20}}{3}.$$

$$\frac{dD}{dt} \Big|_{t=\frac{1}{2}} = \frac{\left(\frac{2}{3} \right) \left(\frac{4}{9} \right) + \left(\frac{4}{3} \right) (4)}{\frac{\sqrt{20}}{3}} = \frac{\frac{8}{27} + \frac{16}{3}}{\frac{\sqrt{20}}{3}} = \frac{152}{27} \cdot \frac{3}{\sqrt{20}} = \frac{152}{9\sqrt{20}}$$

The distance between the particles is increasing at a rate of $\frac{152}{9\sqrt{20}}$.

4-7-7

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the y -axis so that its position at time t is given by $y(t) = t^2 \tan\left(\frac{1}{t}\right)$ for $t > 1$.

(a) Show that the velocity of the particle at time t is given by $v(t) = 2t \tan\left(\frac{1}{t}\right) - \sec^2\left(\frac{1}{t}\right)$ for $t > 1$.

(b) At time $t = \frac{4}{\pi}$, is the particle moving toward the origin or away from the origin? Give a reason for your answer.

(c) The velocity of the particle at time t can be written as $v(t) = \frac{2 \tan\left(\frac{1}{t}\right)}{\frac{1}{t}} - \sec^2\left(\frac{1}{t}\right)$ for $t > 1$.

Find $\lim_{t \rightarrow \infty} v(t)$. Show the work that leads to your answer.

Part A

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 2 points is earned for partial communication of product rule and chain rule with a maximum of one computational error.

The third point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ verification

Solution:

4-7-7

$$v(t) = y'(t) = 2t \tan\left(\frac{1}{t}\right) + t^2 \sec^2\left(\frac{1}{t}\right) \left(-\frac{1}{t^2}\right) = 2t \tan\left(\frac{1}{t}\right) - \sec^2\left(\frac{1}{t}\right)$$

Part B

For the first and second points, trigonometric function values do not need to be evaluated; however, the sign of the expression must be determined correctly to earn the point.

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $y\left(\frac{4}{\pi}\right) > 0$
- ☐ $v\left(\frac{4}{\pi}\right) > 0$
- ☐ answer with reason

Solution:

$$y\left(\frac{4}{\pi}\right) = \left(\frac{4}{\pi}\right)^2 \tan\left(\frac{\pi}{4}\right) = \frac{16}{\pi^2} > 0$$

$$v\left(\frac{4}{\pi}\right) = 2\left(\frac{4}{\pi}\right) \tan\left(\frac{\pi}{4}\right) - \sec^2\left(\frac{\pi}{4}\right) = \frac{8}{\pi} - 2 > 0$$

Because $y\left(\frac{4}{\pi}\right) > 0$, the particle is above the origin at time $t = \frac{4}{\pi}$.

Because $v\left(\frac{4}{\pi}\right) > 0$, the particle is moving up at time $t = \frac{4}{\pi}$.

Therefore, the particle is moving away from the origin at time $t = \frac{4}{\pi}$.

Part C

The first point does not require supporting work.

The second point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing $= \frac{0}{0}$ or not using limit notation) impacts the second point.

The third point does not require a simplified answer. Any differentiation or computation error impacts the third point.

4-7-7

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} \sec^2 \left(\frac{1}{t} \right) = 1$
- ☐ application of L'Hospital's Rule
- ☐ answer

Solution:

$$\lim_{t \rightarrow \infty} 2 \tan \left(\frac{1}{t} \right) = 2 \tan 0 = 0$$

$$\lim_{t \rightarrow \infty} \frac{1}{t} = 0$$

$$\lim_{t \rightarrow \infty} \sec^2 \left(\frac{1}{t} \right) = \sec^2 0 = 1$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow \infty} \frac{2 \tan \left(\frac{1}{t} \right)}{\frac{1}{t}} = \lim_{t \rightarrow \infty} \frac{2 \sec^2 \left(\frac{1}{t} \right) \left(-\frac{1}{t^2} \right)}{-\frac{1}{t^2}} = \lim_{t \rightarrow \infty} 2 \sec^2 \left(\frac{1}{t} \right) = 2$$

Therefore,

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} \left(\frac{2 \tan \left(\frac{1}{t} \right)}{\frac{1}{t}} - \sec^2 \left(\frac{1}{t} \right) \right) = 2 - 1 = 1$$

4-7-7

60. $\lim_{x \rightarrow 0} \frac{\int_0^x \cos(2t) \, dt}{5x}$ is
- (A) 0
- (B) $\frac{1}{5}$
- (C) $\frac{2}{5}$
- (D) nonexistent

**Answer B**

Correct. Let $g(x) = \int_0^x \cos(2t) \, dt$. This function is continuous because it is differentiable with $g'(x) = \cos(2x)$.

Therefore, $\lim_{x \rightarrow 0} \int_0^x \cos(2t) \, dt = \lim_{x \rightarrow 0} g(x) = g(0) = \int_0^0 \cos(2t) \, dt = 0$.

Because $\lim_{x \rightarrow 0} (5x) = 0$ also, $\lim_{x \rightarrow 0} \frac{\int_0^x \cos(2t) \, dt}{5x}$ can be evaluated using L'Hospital's rule.

$$\lim_{x \rightarrow 0} \frac{\int_0^x \cos(2t) \, dt}{5x} = \lim_{x \rightarrow 0} \frac{\cos(2x)}{5} = \frac{\cos(0)}{5} = \frac{1}{5}$$

4-7-7

61.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	-6	7	4	-1
4	5	6	0	5

The table shown gives values of the twice-differentiable functions f and g and their derivatives f' and g' at selected values of x . What is $\lim_{x \rightarrow 2} \frac{g(x^2)}{3x+f(x)}$?

(A) $\frac{1}{2}$

(B) $\frac{20}{13}$

(C) 2



(D) The limit does not exist.

Answer C

Correct. Because f and g are differentiable, they are also continuous. Therefore, $\lim_{x \rightarrow 2} g(x^2) = g(4) = 0$ and $\lim_{x \rightarrow 2} (3x + f(x)) = 6 + f(2) = 0$. The limit is of indeterminate form and can be evaluated using L'Hospital's Rule. Because f' and g' are differentiable, they are also continuous. Therefore,

$$\lim_{x \rightarrow 2} \frac{g(x^2)}{3x+f(x)} = \lim_{x \rightarrow 2} \frac{2xg'(x^2)}{3+f'(x)} = \frac{4g'(4)}{3+f'(2)} = \frac{4(5)}{3+7} = \frac{20}{10} = 2.$$

4-7-7

62. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(3t^2) dt}{4x^3}$ is

(A) $-\frac{1}{4}$

(B) 0

(C) $\frac{1}{4}$

(D) nonexistent

**Answer C**

Correct. Let $g(x) = \int_0^x \sin(3t^2) dt$. This function is continuous because it is differentiable with

$$g'(x) = \sin(3x^2). \text{ Therefore, } \lim_{x \rightarrow 0} \int_0^x \sin(3t^2) dt = \lim_{x \rightarrow 0} g(x) = g(0) = \int_0^0 \sin(3t^2) dt = 0.$$

In addition, $\lim_{x \rightarrow 0} (4x^3) = 0$, so $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(3t^2) dt}{4x^3}$ can be evaluated using L'Hospital's Rule.

$$\lim_{x \rightarrow 0} \frac{\int_0^x \sin(3t^2) dt}{4x^3} = \lim_{x \rightarrow 0} \frac{\sin(3x^2)}{12x^2}$$

This is also an indeterminate form, so L'Hospital's Rule can be used again.

$$\lim_{x \rightarrow 0} \frac{\sin(3x^2)}{12x^2} = \lim_{x \rightarrow 0} \frac{6x \cos(3x^2)}{24x} = \lim_{x \rightarrow 0} \frac{\cos(3x^2)}{4} = \frac{1}{4}$$

4-7-7

63. $\lim_{x \rightarrow \pi/4} \frac{\int_{\pi/4}^x \tan(t) dt}{x^2 - \frac{\pi^2}{16}}$ is

(A) 0

(B) $\frac{2}{\pi}$ (C) $\frac{4}{\pi}$

(D) nonexistent

**Answer B**

Correct. Because $f(x) = \int_{\pi/4}^x \tan(t) dt$ is differentiable for $0 < x < \frac{\pi}{2}$ with $f'(x) = \tan(x)$, the function f is continuous for $0 < x < \frac{\pi}{2}$. Therefore $\lim_{x \rightarrow \pi/4} \int_{\pi/4}^x \tan(t) dt = \lim_{x \rightarrow \pi/4} f(x) = f\left(\frac{\pi}{4}\right) = \int_{\pi/4}^{\pi/4} \tan(t) dt = 0$.

Moreover, $\lim_{x \rightarrow \pi/4} \left(x^2 - \frac{\pi^2}{16}\right) = \left(\frac{\pi^2}{16} - \frac{\pi^2}{16}\right) = 0$, and so L'Hospital's Rule can be used to find the limit of this indeterminate form, as follows.

$$\lim_{x \rightarrow \pi/4} \frac{\int_{\pi/4}^x \tan(t) dt}{x^2 - \frac{\pi^2}{16}} = \lim_{x \rightarrow \pi/4} \frac{\frac{d}{dx} \int_{\pi/4}^x \tan(t) dt}{\frac{d}{dx} \left(x^2 - \frac{\pi^2}{16}\right)} = \lim_{x \rightarrow \pi/4} \frac{\tan(x)}{2x} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

64. Let g be the function given by $g(x) = \int_3^x (t^2 - 5t - 14) dt$. What is the x -coordinate of the point of inflection of the graph of g ?

(A) -2

(B) $\frac{5}{2}$

(C) 3

(D) 7

**Answer B**

Correct. To find the point of inflection of the graph of g , determine where g'' changes sign.

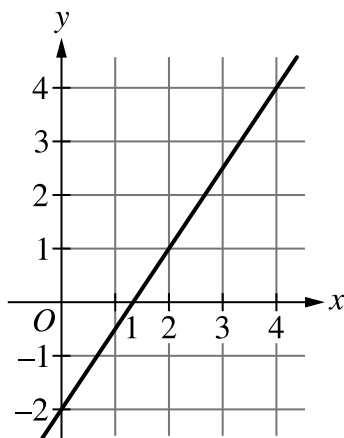
$$g'(x) = x^2 - 5x - 14$$

$$g''(x) = 2x - 5$$

Then $g''(x) = 0$ at $x = \frac{5}{2}$. Since $g''(x) < 0$ for $x < \frac{5}{2}$ and $g''(x) > 0$ for $x > \frac{5}{2}$, the graph of g changes concavity at $x = \frac{5}{2}$ and therefore, the graph of g has a point of inflection at $x = \frac{5}{2}$.

4-7-7

65.

Graph of f

The graph of the function f is shown. The value of $\lim_{x \rightarrow 2} \frac{f(x)-1}{\sin(\frac{\pi}{2}x)}$ is

(A) $-\frac{3}{2}$

(B) $-\frac{3}{\pi}$

(C) $-\frac{1}{\pi}$

(D) nonexistent

**Answer B**

Correct. Since $\lim_{x \rightarrow 2} (f(x) - 1) = f(2) - 1 = 1 - 1 = 0$ and $\lim_{x \rightarrow 2} \sin\left(\frac{\pi}{2}x\right) = \sin(\pi) = 0$, the limit can be evaluated using L'Hospital's Rule.

$$\lim_{x \rightarrow 2} \frac{f(x)-1}{\sin(\frac{\pi}{2}x)} = \lim_{x \rightarrow 2} \frac{f'(x)}{\frac{\pi}{2}\cos(\frac{\pi}{2}x)} = \frac{\frac{3}{2}}{\frac{-\pi}{2}} = -\frac{3}{\pi}$$